

SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that seven open unit equilateral triangles cannot cover a regular unit hexagon. The center and six vertices force distinct covering triangles. The key obstruction is a tradeoff between boundary coverage and interior coverage: a vertex triangle covering more than one unit of incident boundary misses its own radial midpoint. Boundary gaps and midpoint coverage then reduce the proof to length budgets, area loss, and fixed finite-enclosure obstructions. The latter use five points for a selected gap, four for a supported midpoint, and nine when the vertex triangles cover the boundary. All local geometric calculations are included. In the nine-point argument, the final disk-tangent inequalities are verified by an exact polynomial certificate over $\mathbb{Q}(\sqrt{3})$.

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1. INTRODUCTION AND PROOF OUTLINE

1.1. The geometric obstruction. We prove that seven open equilateral triangles of side one cannot cover a regular hexagon of side one. The obstruction is not a shortage of total area: the hexagon has the area of six unit triangles. Instead, the same triangles must cover both the boundary and the interior, and these two obligations compete.

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Seven triangles, seven prescribed points. The center and the six vertices of the hexagon are pairwise at distance at least one. An open unit triangle cannot contain two of them. Thus any hypothetical cover assigns one triangle to the center and one to each vertex. We call these the C triangle and the six V triangles. This elementary observation is the starting point of the proof, not merely a labeling convention; its precise statement is lemma 2.3.

Boundary reach has an interior cost. Call a V triangle *supercritical* when the sum of the lengths it covers on its two incident hexagon edges is greater than one. If the V triangles cover the entire boundary, neighboring open traces must overlap. Their total length is therefore greater than six, so at least one V triangle is supercritical. The crucial local fact is that a supercritical V triangle misses the midpoint of the segment from its vertex to the center (lemma 2.14). Extra boundary reach creates an interior obligation for another triangle.

Two branches of the argument. If the V triangles cover the entire boundary, two or more supercritical triangles waste too much area outside the hexagon. Exactly one leads instead to nine points that the C triangle must contain, but that cannot fit inside an open unit triangle. If the V triangles leave a boundary gap, the C triangle must cover it. Reaching the boundary restricts the C triangle to exactly one radial midpoint. Length budgets then leave only a small number of ways to cover the other necessary midpoints. These are excluded by five-point or four-point enclosures, a perimeter estimate, or a replacement that preserves the boundary and radial segments. The detailed hypotheses are given in Table 1; this paragraph describes the logic before introducing the classification notation.

The proof therefore has three geometric measurements: length on a fixed network, area lost outside the hexagon, and the least equilateral triangle containing a fixed set of points. The local calculations serve these three measurements. The exact computer-assisted step is confined to the final zero-gap enclosure argument; it is not a numerical search over covers.

1.2. Notation, theorem, and canonical labeling. All distances, angles, and areas are Euclidean. A unit equilateral triangle has side length one; its open version is its interior. We write $[x, y]$ for the closed segment, $\text{conv}K$ for the convex hull, and $\text{int}K$ and ∂K for the interior and boundary. The relative interior $\text{relint}K$ is its interior within its affine span. The measure $\mathcal{H}^1$ is one-dimensional Hausdorff measure, which agrees with length on segments. We use $[t]_+ = \max\{t, 0\}$ and $\text{cross}(x, y) = x_1y_2 - x_2y_1$ for vectors in $\mathbb{R}^2$. A *trace* on a set E means its intersection with E .

Let

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right), \quad H = \text{conv}\{V_0, \dots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \quad r_i = [O, V_i], \quad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^5 r_i.$$

We call S the *hexagon skeleton*. For $X \in \mathbb{R}^2$ and $r \geq 0$, write $\mathbb{B}(X, r) = \{Y : \|Y - X\| \leq r\}$ for the closed disk.

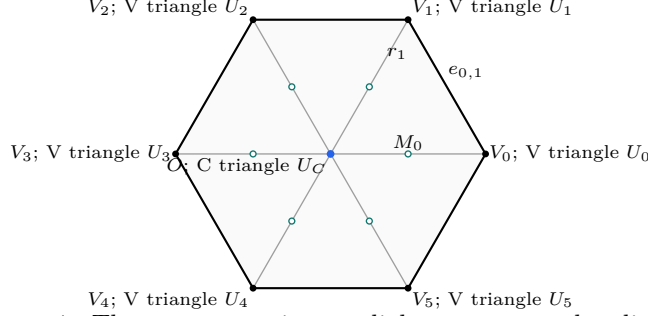


FIGURE 1. The center, vertices, radial segments, and radial mid-points of H .

For a nonempty compact set $K \subset \mathbb{R}^2$, define its *equilateral enclosure side* by

$$(1) \quad \Lambda(K) := \inf\{s > 0 : K \subset T \text{ for a closed equilateral triangle } T \text{ of side } s\}.$$

Then

$$(2) \quad \mathcal{H}^1(\partial H) = 6, \quad \mathcal{H}^1\left(\bigcup_{i=0}^5 r_i\right) = 6, \quad \mathcal{H}^1(S) = 12.$$

Theorem 1.1 (Main theorem). *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem). *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

By lemma 2.3, a hypothetical cover has a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \quad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle at V_i* . Set

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

A *role* means one of these labeled triangles. Statements about an arrangement of roles assume $O \in U_C$ and $V_i \in U_i$; coverage of H , S , or ∂H is an additional hypothesis, stated when needed. A *skeleton cover* covers S , not necessarily all of H . An edge is *center-free* when it is disjoint from U_C .

Which part of the hexagon must remain covered? A full cover also covers the skeleton S . The converse is not assumed. We retain this weaker target because one later replacement preserves the skeleton, but need not preserve the interior of H . Statements about a skeleton cover are therefore needed for the logic of the replacement, not just for estimating lengths. The open sets U_i determine coverage; their closures T_i are used for maxima, incidence classifications, and upper bounds. A point of $T_i \setminus U_i$ is not covered by U_i .

1.3. How to read the proof. Section 2 introduces the incidence classes, actual reaches, gaps, and midpoint restrictions, in that order. Its final routing table gives the exhaustive split after the relevant notation is available. Sections 3 and 4 establish the length and area budgets. Section 5 explains how to fix witnesses before testing an arbitrary enclosing triangle; the selected-gap construction is worked through point by point. Section 6 assembles the obstructions.

The main text states the geometric inputs and explains their use. Appendices A–E retain the complete local calculations, including both replacement charts. Appendix F contains the exact tangent-residual certificate and identifies its accompanying finite data and replay programs. Reading a calculation can be postponed, but none is being replaced by an illustration or an experimental assertion.

2. COMMON GEOMETRY AND STRUCTURAL REDUCTIONS

We prove the structural facts used by all three mechanisms. The coordinate calculations are in Appendix A.

2.1. Open and closed formulations.

Lemma 2.1 (Compact covering margin). *If a nonempty compact set K is covered by finitely many open unit equilateral triangles U_ν , then moving every side line inward by the same sufficiently small positive distance gives a cover by closed equilateral triangles of side less than one. In particular, $K \subset U$ implies $\Lambda(K) < 1$, with Λ defined in (1).*

Proof. The continuous function $m(x) = \max_\nu \text{dist}(x, \mathbb{R}^2 \setminus U_\nu)$ has positive minimum η on K . Move every side line inward by $0 < \rho < \min\{\eta, \sqrt{3}/6\}$. The resulting closed triangles have side $1 - 2\sqrt{3}\rho$, and each $x \in K$ remains in a triangle realizing $m(x) \geq \eta > \rho$. Conversely, if $\Lambda(K) < 1$, choose a closed enclosure of side below one and enlarge it about its incenter to side one. The smaller closed triangle lies in the interior of the enlarged triangle. $\square$

Proposition 2.2 (Compactness and scaling equivalence). *The following assertions are equivalent.*

- (1) H is covered by seven open equilateral triangles of side 1.
- (2) For some $\varepsilon > 0$, H is covered by seven closed equilateral triangles of side $1 - \varepsilon$.
- (3) For some $L > 1$, $H_L := LH$ is covered by seven closed equilateral triangles of side 1.

Proof. Apply lemma 2.1 to obtain (1) $\Rightarrow$ (2). Enlarging the smaller closed triangles about their incenters gives (2) $\Rightarrow$ (1); dilation by $(1 - \varepsilon)^{-1}$ identifies (2) and (3). $\square$

Lemma 2.3 (Distinct triangles). *The seven triangles can be labelled so that*

$$O \in U_C, \quad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven triangles are distinct. Consequently

$$O \in \text{int}(T_C), \quad V_i \in \text{int}(T_i).$$

Proof. The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least 1, whereas two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Selecting a covering triangle for each point therefore gives a bijection. $\square$

2.2. C triangle and V triangle classifications. Set

$$N_E(T_C) := |\{i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0\}|.$$

Proposition 2.4 gives $N_E(T_C) \in \{0, 1, 2\}$; we call these types CE0, CE1, CE2, respectively.

Point contacts have zero trace length, so they do not create an additional positive boundary trace in the definition of $N_E(T_C)$.

Proposition 2.4 (Exhaustive center types). *The closed C triangle T_C has exactly one of the types CE0, CE1, CE2; equivalently,*

$$N_E(T_C) \leq 2.$$

Proof. If $N_E(T_C) \geq 3$, two of the traced edges are nonadjacent in the boundary six-cycle. Choose a relative-interior point from each positive-length trace. By lemma A.1, those points are more than unit distance apart, so a unit equilateral triangle cannot contain both. $\square$

For the V triangle T_i , let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|,$$

with indices modulo 6. The raw pair $(3, 0)$ says that all three vertices of T_i lie outside H ; it does not say that T_i is disjoint from H , since $V_i \in \text{int}(T_i) \cap H$.

Lemma 2.5 (Local wedge). *Every original V triangle has $1 \leq o(T_i) \leq 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one vertex of T_i belongs to H ; if both adjacent radial traces have positive length, at least two vertices belong to H .*

Proof. This is the wedge-incidence calculation in lemma A.3. The incidence statement concerns the closures; openness is used separately when forcing points. $\square$

Raw roles with $(o(T_i), n(T_i)) = (3, 0)$ admit an exact-trace normalization.

Why normalize before measuring reaches? The raw case with all three triangle vertices outside H is an incidence artifact: a translation can leave both its open and closed traces in H unchanged. The next normalization removes that extra case before any actual reach is defined. It neither enlarges the covered part of H nor changes which boundary points are covered. This differs from the trace majorant introduced later, which is not a new open covering triangle.

Proposition 2.6 (Exact-trace normalization of raw $(3, 0)$ roles). *Let U be an original open V triangle at V_i , put $T = \overline{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a same-orientation translate $\widehat{U}$, with closure $\widehat{T}$, such that*

$$(3) \quad V_i \in \widehat{U}, \quad (o(\widehat{T}), n(\widehat{T})) = (2, 0),$$

and

$$(4) \quad T \cap H = \widehat{T} \cap H, \quad U \cap H = \widehat{U} \cap H.$$

In particular, all three actual maximal reaches, and whether the two boundary reaches have sum greater than one, are unchanged.

Proof. The two-orientation affine calculation and the trace-preserving translation are given in lemma A.4. $\square$

We apply this translation simultaneously and reuse U_i, T_i for the normalized triangles.

The normalized exhaustive types used below are

$(o(T_i), n(T_i))$	established type
(1, 0)	Vd0
(2, 0)	Vd0
(1, 1)	Vd1
(1, 2)	Vd2
(2, 1)	T3-like.

Proposition 2.7 (Exhaustive normalized vertex types). *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, T3-like in the displayed type dictionary.*

Proof. Lemma 2.5 gives $o \leq 2$ when $n \geq 1$ and $o \leq 1$ when $n = 2$, while $o \geq 1$. Proposition 2.6 replaces the only additional raw pair, $(3, 0)$, by $(2, 0)$. $\square$

The next construction is a trace majorant, not a new open covering triangle.

Proposition 2.8 (T3-like closed-trace domination). *Let T be the closure of an original T3-like V triangle at V_i . There is a same-orientation translate $\hat{T}$ such that*

$$V_i \in \text{relint}(a \text{ side of } \hat{T}), \quad T \cap H \subsetneq \hat{T} \cap H,$$

and $(o(\hat{T}), n(\hat{T})) = (2, 1)$, while

$$\text{int}(T) \cap (H \setminus \{V_i\}) \subseteq \text{int}(\hat{T}).$$

Since V_i becomes a boundary point, $\hat{T}$ is only a closed-trace majorant.

Proof. See the two-orientation feasibility calculation in lemma A.5. $\square$

2.3. Initial reaches and boundary gaps. The actual maximal reaches are

$$\begin{aligned} A_i &:= \max\{x \in [0, 1] : V_i + x(V_{i-1} - V_i) \in T_i\}, \\ B_i &:= \max\{x \in [0, 1] : V_i + x(V_{i+1} - V_i) \in T_i\}, \\ C_i &:= \max\{x \in [0, 1] : V_i + x(O - V_i) \in T_i\}. \end{aligned}$$

The same numbers are the suprema of the corresponding reaches in the open triangle U_i . The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and *nonsupercritical* when $A_i + B_i \leq 1$. Put

$$(5) \quad N_+ := |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|.$$

Lowercase reaches are used only after a selection satisfying

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i, \quad 0 \leq c_i \leq C_i.$$

They never define N_+ .

Lemma 2.9 (Initial reach segments). *On the three segments from V_i toward V_{i-1}, V_{i+1}, O , the closed parameter sets are respectively*

$$[0, A_i], \quad [0, B_i], \quad [0, C_i],$$

and the open parameter sets are

$$[0, A_i), \quad [0, B_i), \quad [0, C_i).$$

Thus the closed maxima equal the corresponding open-triangle suprema.

Proof. The other endpoint of each segment is a distinguished point at distance one from V_i , and is not even in T_i : otherwise moving V_i slightly away from it inside U_i would exceed diameter one. Thus $0 < A_i, B_i, C_i < 1$. Each intersection with a segment issuing from V_i is convex. Closedness of T_i , openness of U_i , and $T_i = \overline{U_i}$ give the stated endpoints. $\square$

Parametrize $e_{i,i+1}$ from V_i to V_{i+1} by

$$X_i(x) := V_i + x(V_{i+1} - V_i), \quad 0 \leq x \leq 1.$$

Define the *boundary gap*

$$J_i := e_{i,i+1} \setminus (U_i \cup U_{i+1}), \quad N_{\text{gap}} := |\{i : J_i \neq \emptyset\}|.$$

Lemma 2.10 gives $J_i = X_i([B_i, 1 - A_{i+1}])$ when $B_i + A_{i+1} \leq 1$, and $J_i = \emptyset$ otherwise. Equality is a singleton gap, not an overlap.

Lemma 2.10 (Exhaustiveness of the gap split). *One has*

$$J_i = \begin{cases} X_i([B_i, 1 - A_{i+1}]), & B_i + A_{i+1} \leq 1, \\ \emptyset, & B_i + A_{i+1} > 1. \end{cases}$$

Every nonempty gap, including a singleton, is missed by all open V triangles. Under a hypothetical cover it lies in U_C and creates a positive-length C triangle trace. Consequently a closed C triangle T_C of type CE0, CE1, CE2 permits respectively at most 0, 1, 2 gaps, and

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Proof. By lemma 2.9, the incident open traces have parameters $[0, B_i]$ and $[1 - A_{i+1}, 1]$. Their complement is the stated closed interval. A nonincident V triangle cannot contain a relative-interior point of the edge: that point is more than unit distance from its distinguished vertex. Hence a gap lies in U_C ; openness turns even a singleton V-gap into a positive-length C trace. Proposition 2.4 gives the count and the final equivalence. $\square$

Why the threshold is one. On a gap-free edge, the open traces overlap, so $B_i + A_{i+1} > 1$. If every edge is gap-free, summing gives

$$6 < \sum_{i=0}^5 (B_i + A_{i+1}) = \sum_{i=0}^5 (A_i + B_i).$$

At least one V triangle must therefore be supercritical. A nonempty gap breaks this cyclic argument: it is precisely where the C triangle takes over a boundary obligation. Equality $B_i + A_{i+1} = 1$ still leaves the common endpoint uncovered by the two open V triangles; it belongs to the gap branch, not to the overlap branch.

The distinction between actual and selected reaches is also geometric. The numbers A_i, B_i, C_i describe a fixed triangle. The later lower bounds a_i, b_i, c_i prescribe points it must contain and may be reduced when making a common comparison. Reducing a demand does not change the original triangle or the count N_+ .

Lemma 2.11 (T3-like V triangles are nonsupercritical). *Every original T3-like V triangle satisfies*

$$(6) \quad A_i + B_i \leq 1.$$

Moreover $M_i \notin T_i$. In particular, a T3-like role is not counted by N_+ and cannot supply its own midpoint.

Proof. Both assertions follow from the Type-II calculation in lemma A.5. $\square$

A V role has *positive adjacent-radial support*, or simply *positive support*, when $n(T_i) > 0$. Put

$$N_{\text{sp}} := |\{i : n(T_i) > 0\}|.$$

Positive-support roles are nonsupercritical, so the number of roles that are supercritical or have positive adjacent-radial support is $N_+ + N_{\text{sp}}$. The skeleton budget excludes nonzero-gap covers with $N_+ + N_{\text{sp}} \geq 3$.

2.4. Strict handoffs and midpoint forcing.

Proposition 2.12 (Strict boundary handoffs). *Assume that $U_0, \dots, U_5$ cover ∂H . There are choices*

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1}, \quad b_i := \xi_i,$$

such that

$$0 < a_i < A_i, \quad 0 < b_i < B_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

The selected anchors lie in U_i . If exactly one actual V triangle is supercritical, every strict selection has that same unique selected supercritical index. If at least two actual V triangles are supercritical, some simultaneous strict selection retains at least two.

Proof. The overlap intervals and the cyclic selector are proved in lemma A.6. Geometrically, each ξ_i lies in the overlap of two adjacent open edge traces, which makes both associated reach demands strict. $\square$

Lemma 2.13 (Boundary deficit and overlap identities). *For the actual reaches put $s_i = 1 - A_i - B_i$ and $\omega_i = B_i + A_{i+1} - 1$. Then*

$$A_{i+1} - A_i = s_i + \omega_i, \quad B_i - B_{i+1} = s_{i+1} + \omega_i.$$

On a nonsupercritical gap-free path the A_i increase and the B_i decrease strictly. Weak handoffs give the weak version. Around the full cycle, $\sum_i \omega_i = -\sum_i s_i$.

Proof. Expand the definitions and telescope. Nonsupercriticality gives $s_i \geq 0$, and a gap-free edge has $\omega_i > 0$ because its incident traces are open. A supercritical role has $s_i < 0$; propagation through it is not licensed. $\square$

Lemma 2.14 (Self-midpoint obstruction). *Let u, v be unit vectors meeting at 120° , and let $a, b \in [0, 1]$. If a closed unit equilateral triangle contains*

$$0, \quad au, \quad bv, \quad \frac{u+v}{2},$$

then $a + b \leq 1$. Consequently

$$M_i \in T_i \implies A_i + B_i \leq 1.$$

If the boundary points and midpoint are contained with positive margin, the conclusion is strict.

Proof. This is the admissible-set corollary lemma A.8. $\square$

Proposition 2.15 (CE1/CE2 exactly-one-midpoint property). *If $O \in \text{int}(T_C)$ and T_C is CE1 or CE2, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

This midpoint belongs to U_C . Both possible positive boundary traces are on the two hexagon edges incident to the same vertex.

Proof. Normalize one positive boundary trace. The signed-center optimization proposition A.10 gives the exact midpoint set $\{M_0\}$, with radial exit $d_0^C > 1/2$ and positive side slacks at M_0 . Its boundary traces are confined to $e_{5,0}, e_{0,1}$. Undoing the symmetry proves the assertions. $\square$

The interior hypothesis is essential: $\text{conv}\{O, V_0, V_1\}$ has a boundary edge overlap and contains both M_0 and M_1 , but O is not interior to it.

2.5. Exhaustive routing. When the C triangle has type CE1 or CE2, proposition 2.15 shows that it contains exactly one of $M_0, \dots, M_5$; this is the *C midpoint*. If it is M_k , the role at V_k is called *center-based*; a supercritical role there is *center-aligned*, and a role at another vertex is *away* or *off-center*. A V role that contains an adjacent midpoint openly is called its *supplier* or *rescuer*. A *gap-free nonsupercritical path* is a consecutive sequence of nonsupercritical V roles whose shared edges have empty gaps.

We use BC for the selected-gap five-point construction, D for the supported-midpoint four-point construction, and F for the zero-gap nine-point construction, all defined in Section 5. N0 denotes theorem 5.14, the assertion that every skeleton cover has a supercritical V role.

Zero gaps mean that the V triangles cover ∂H , and the proof splits according to N_+ . With a gap, the C triangle is CE1 or CE2; N0 and the skeleton budget leave one supercritical role and at most one positive-support role. Its position relative to the C midpoint determines the remaining obstruction.

N_{gap}	N_+	Additional condition	Closing mechanism
0	0	Arbitrary V types	Strict overlap
0	1	Arbitrary V types	Nine-point F
0	≥ 2	Arbitrary V types	Cyclic area loss
≥ 1	0	Arbitrary V types	N0, through BC
≥ 1	≥ 1	$N_+ \geq 2$ or $N_+ + N_{\text{sp}} \geq 3$	Skeleton budget
≥ 1	1	$N_{\text{sp}} \leq 1$	Midpoint supplier: BC, D, replacement, or Vd2 length

TABLE 1. Disjoint routes for the original open cover. Nonzero gaps force CE1 or CE2; both classes share the same placement reduction. Replacement preserves the skeleton and ends at N0, without preserving number of gaps.

The rows are disjoint and exhaustive: separate zero from nonzero gaps, then $N_+ = 0$, and finally the displayed high-count condition. Lemma 5.15 resolves the

last row. BC uses the same selected-gap witness and the same finite-caliper criterion in CE1 and CE2.

3. LENGTH BUDGETS ON THE BOUNDARY AND SKELETON

The target is a network of known total length, not the whole two-dimensional hexagon. Each triangle has a local upper bound on the length it can cover. Because the covering sets are open, some covering pieces must overlap; hence the sum of their lengths must be *strictly greater* than the target length. An upper bound equal to that target is already a contradiction.

For the skeleton, an ordinary nonsupercritical V triangle without adjacent radial support contributes at most two. Supercritical V triangles and those with adjacent radial support each contribute less than $3/2$. When the C triangle reaches the boundary, its skeleton contribution is also below $3/2$. The count argument below simply adds these local budgets. The local caps require their stated hypotheses; no unrestricted skeleton cap is being asserted for the C triangle.

For a closed triangle T and a rectifiable target E , define

$$L_E(T) := \mathcal{H}^1(T \cap E).$$

Lemma 3.1 (Connected open-cover budget). *Let X be compact and connected, with a finite Borel measure μ positive on every nonempty relatively open subset. Let $U_1, \dots, U_m$ be open sets covering X , put $T_j = \overline{U_j}$, and assume at least two of the relative pieces $U_j \cap X$ are nonempty. Then*

$$\mu(X) < \sum_j \mu(U_j \cap X) \leq \sum_j \mu(T_j \cap X).$$

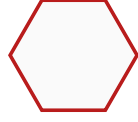
Consequently closed-piece upper bounds whose sum is at most $\mu(X)$ already contradict the open cover.

Proof. If the nonempty relative pieces were pairwise disjoint, one and the union of the others would disconnect X . Some pair therefore overlaps in a nonempty relatively open set, of positive measure. Integrating the covering multiplicity gives strict inequality. Closure only enlarges each piece. $\square$

For $X = \partial H$ or S , use length; for $X = H$, use area. Each measure has the stated full-support property. The six V roles meet each target in positive measure; the C role does so on H and S , but need not meet ∂H . In particular,

$$(7) \quad \mathcal{H}^1(E) < L_E(T_C) + \sum_{i=0}^5 L_E(T_i), \quad E \in \{\partial H, S\},$$

whenever these targets are covered. Empty center boundary traces are simply omitted from the collection. The local geometric caps remain separate from this common final budget argument.

 ∂H : length 6 S : length 12

the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 2. The perimeter and full-skeleton targets. The skeleton is used only under the hypotheses of the corresponding trace budget.

3.1. Skeleton bounds.

Lemma 3.2 (Center skeleton cap). *If $N_E(T_C) \in \{1, 2\}$ and*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

then

$$L_S(T_C) < \frac{3}{2}.$$

Proof. This is the signed-center trace optimization lemma B.4. $\square$

Lemma 3.3 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle. If*

$$\exists j \in \{i-1, i+1\} \quad \mathcal{H}^1(T_i \cap r_j) > 0,$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0 . The five components of S on which a unit triangle containing V_0 can have positive trace become the corresponding local components of the translated skeleton. The positive adjacent-radial trace becomes a positive boundary trace there. Proposition 2.15 and Lemma 3.2 give the bound; every nonlocal component is more than unit distance from V_0 , apart from zero-measure endpoints. $\square$

Lemma 3.4 (No-support skeleton cap). *If*

$$A_i + B_i \leq 1,$$

$$\mathcal{H}^1(T_i \cap r_{i-1}) = 0, \quad \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \leq 2.$$

Proof. For $s > 0$ and $j \notin \{i-1, i, i+1\}$, $\|V_i - sV_j\|^2 = 1 + s^2 - 2s \cos((j-i)\pi/3) > 1$. Thus only the incident boundary traces and own radial trace contribute; their total is $A_i + B_i + C_i \leq 2$. $\square$

Lemma 3.5 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. The exact admissible-cell and polynomial-sign calculation is lemma B.5. $\square$

Lemma 3.6 (Positive-support rescuer). *Assume that the open roles cover S , that $N_E(T_C) \in \{1, 2\}$, and, after symmetry,*

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

If $N_+ \geq 2$, then for a supercritical index $s \neq 0$ there is $j \in \{s-1, s+1\}$ such that

$$M_s \in U_j, \quad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Proof. Choose a supercritical $s \neq 0$. Lemma 2.14 gives $M_s \notin T_s$, and the displayed midpoint identity gives $M_s \notin T_C$. The open cover therefore puts M_s in another U_j . Diameter locality restricts j to $s-1$ or $s+1$, and openness makes its trace on r_s have positive length. $\square$

Theorem 3.7 (Direct CE1/CE2 skeleton count). *If*

$$N_E(T_C) \in \{1, 2\},$$

$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

and

$$(8) \quad N_+ + N_{\text{sp}} \geq 3,$$

then the seven triangles do not cover S .

Proof. By proposition B.1 and lemma 2.11, every positive-support role is nonsupercritical. Hence the two index sets are disjoint. With $k = N_+ + N_{\text{sp}}$, the preceding caps give

$$L_S(T_C) + \sum_i L_S(T_i) < \frac{3}{2} + \frac{3}{2}k + 2(6-k) = \frac{27-k}{2} \leq 12,$$

contrary to (7). $\square$

3.2. Boundary trace bounds. The following bounds suffice for the perimeter exit.

Theorem 3.8 (Boundary trace bounds). *For closures of the original open roles,*

$$L_{\partial H}(T_C) = 0 \quad (\text{CE0}), \quad L_{\partial H}(T_C) < \frac{1}{2} \quad (\text{CE1 or CE2}),$$

and

$$(9) \quad L_{\partial H}(T_i) = A_i + B_i \leq \frac{2}{\sqrt{3}}.$$

A nonsupercritical role contributes at most 1; a Vd1 or Vd2 role contributes less than 1/2.

Proof. Boundary locality and the diameter bound give $A_i^2 + A_i B_i + B_i^2 \leq 1$, hence $3(A_i + B_i)^2/4 \leq 1$. The remaining caps are corollary A.11 and proposition B.1. $\square$

Proposition 3.9 (Vd2 adjacent-midpoint branch). *Suppose the C triangle is CE1 or CE2 and $N_+ = 1$. If a Vd2 role contains an adjacent midpoint, the perimeter is not covered.*

Proof. By lemma B.3, that role contributes strictly less than 1/3. The common boundary caps give

$$L_{\partial H}(T_C) + \sum_i L_{\partial H}(T_i) < \frac{1}{2} + \frac{2}{\sqrt{3}} + \frac{1}{3} + 4 < 6.$$

Apply lemma 3.1. $\square$

4. AREA-LOSS ESTIMATE

The area-loss method records how much of each V triangle must lie outside H . The local estimate applies to every V triangle. A final cyclic argument adds these losses without imposing a center type.

Normalize area by $A_\Delta = \sqrt{3}/4$ and write

$$\mathcal{L}(T) := \frac{\text{area}(T \setminus H)}{A_\Delta}.$$

The hexagon has normalized area six, and a unit equilateral triangle has normalized area one.

There is exactly one normalized unit of excess area: seven triangles are available for a hexagon of area six. Thus a total loss greater than one outside H makes a cover impossible, even before accounting for overlap inside H . The local estimate quantifies the loss forced by a V triangle's two boundary anchors. The cyclic step is used only when the V triangles cover the whole boundary; it links their anchors through the strict handoffs. The area argument does not assume a particular C triangle type.

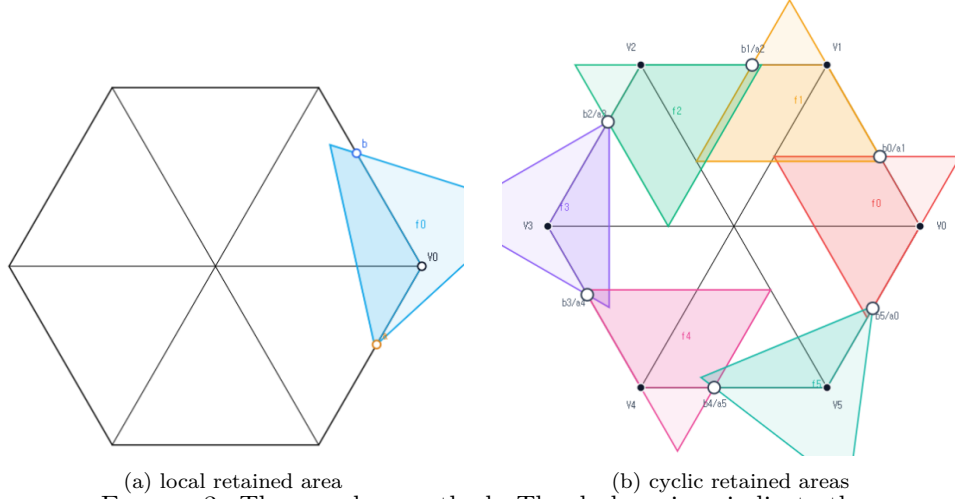


FIGURE 3. The area-loss method. The dark regions indicate the portions retained in H ; the loss $\mathcal{L}(T)$ is the area outside H .

4.1. Local area loss.

Theorem 4.1 (Local square loss). *Let T be a closed unit equilateral triangle containing V_i , $V_i + a(V_{i-1} - V_i)$, and $V_i + b(V_{i+1} - V_i)$, where $a, b \in [0, 1]$. Then*

$$(10) \quad \mathcal{L}(T) \geq \min\{a, b\}^2,$$

and, if $a + b > 1$,

$$(11) \quad \mathcal{L}(T) \geq \max\{a, b\}^2.$$

Proof. The two exhaustive orientation families give these lower bounds. Their polygon areas and the factorizations proving both inequalities are computed in lemma C.1. $\square$

4.2. Cyclic accumulation. Assume here that the six V triangles cover ∂H , and choose strict handoffs ξ_i, a_i, b_i as in proposition 2.12. An *ascent* at index i means $\xi_{i-1} < \xi_i$, equivalently $a_i + b_i > 1$.

Lemma 4.2 (Cyclic area-loss certificate). *The six actual V triangles satisfy*

$$|\{i \in \mathbb{Z}/6\mathbb{Z} : a_i + b_i > 1\}| \geq 2 \implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

Proof. Put $m = \min_i \xi_i$ and $M = \max_i \xi_i$. The reflection $y_i = 1 - \xi_{-i-1}$ swaps each V triangle's two coordinates and preserves the number of ascents and the total loss. Hence assume $m \leq 1 - M$. Every selected V triangle coordinate is then at least m , so (10) gives loss at least m^2 for every V triangle.

Choose an ascent out of a minimum plateau, $\xi_{p-1} = m < \xi_p$. Its V triangle contains the coordinate $1 - m$ and loses at least $(1 - m)^2$ by (11). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$. The other four V triangles lose at least m^2 each, so the total loss is strictly larger than

$$4m^2 + (1 - m)^2 + \frac{1}{4} = 5 \left(m - \frac{1}{5} \right)^2 + \frac{21}{20} > 1.$$

□

Proposition 4.3 (Branch closed by area). *The branch $N_{\text{gap}} = 0$, $N_+ \geq 2$ is impossible.*

Proof. The six V triangles cover ∂H . The at-least-two clause of proposition 2.12 supplies the selection for lemma 4.2. Thus the V triangles have total normalized inside area below five. The C triangle contributes at most one, whereas H has normalized area six. Apply lemma 3.1 with area. □

5. FINITE-ENCLOSURE OBSTRUCTIONS

The finite-enclosure method identifies a compact set that the C triangle must contain, then proves that this set does not fit inside any open unit equilateral triangle.

Put

$$h := \frac{\sqrt{3}}{2}.$$

The closed-disk notation and Λ are defined in Section 1. By lemma 2.1, a compact set contained in an open unit triangle has enclosure side below one. Conversely, $\Lambda(K) < 1$ gives such an open unit enclosure by slightly enlarging a smaller closed enclosure. Thus it is enough to exclude every open unit candidate containing the same fixed set K .

5.1. The two tasks in a finite-enclosure proof. Fix the original seven triangles and call their arrangement $\mathcal{U}$. A nonempty compact witness set $K = K(\mathcal{U})$ is then chosen once and for all. There are two separate tasks:

$$\underbrace{K \subset U_C}_{\text{forced containment}} \quad \text{and} \quad \underbrace{\Lambda(K) \geq 1}_{\text{enclosure obstruction}}.$$

The first makes $\Lambda(K) < 1$ by the compact covering margin, contradicting the second. Most witnesses are points missed by all open V triangles. A structural anchor, such

as the C midpoint, may instead be included because it is already known to belong to U_C ; such an anchor need not be missed by every V triangle.

To prove the second task, suppose an *arbitrary* open unit triangle U' contains the fixed K . Its orientation, offsets, and radial exits may vary. The original reaches and the witness coordinates do not. In particular, U' need not cover the original boundary gaps or complete the original skeleton cover. Any inequality obtained from original coverage must be established before U' is introduced and retained as a fixed input.

Object	What is fixed or required
Original arrangement $\mathcal{U}$	Its actual reaches and all coverage hypotheses of the current case.
Witness set $K(\mathcal{U})$	Chosen from the original arrangement; never recomputed from the candidate.
Candidate U'	Required only to contain K ; its own geometric parameters are recomputed.
Replacement arrangement	A different operation, used later to preserve skeleton coverage, not an enclosure candidate.

The five-point construction uses actual gap endpoints and safely clipped radial capacity bounds. The four-point construction uses a supported radial endpoint and boundary anchors. The zero-gap construction instead uses uniform capacity bounds to choose six radial points and three frontier points. Their common feature is fixed containment, not a common coordinate formula.

5.2. The enclosure calculus: finitely many orientations. The next four results are the general enclosure tools, rather than case-specific estimates. The gauge removes translation and scale; the support-cell argument reduces the remaining orientation variable to a finite list. They do not construct the witnesses or prove that those witnesses are forced into the C triangle.

Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact $K \subset \mathbb{R}^2$, put

$$h_K(n) := \max_{x \in K} \langle x, n \rangle.$$

Proposition 5.1 (Equilateral enclosure gauge). *The least side length $\Lambda(K)$ of a closed equilateral triangle containing K is*

$$(12) \quad \Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^2 h_K(R^j n).$$

It is monotone under inclusion, translation invariant, and positively homogeneous.

Proof. For fixed outward normals n, Rn, R^2n , the least containing triangle has the corresponding three support numbers, and its side is $2/\sqrt{3}$ times their sum. Translation invariance follows from

$$n + Rn + R^2n = 0,$$

while monotonicity and homogeneity are immediate. $\square$

Lemma 5.2 (Support-cell rotation). *A least equilateral enclosure of a nondegenerate polygon can be chosen with one side containing a polygon edge. For a centered disk together with finitely many points, it suffices to test point–point and point–disk*

ties, or a disk-only orientation. Here a support cell is an open angular interval on which the maximizing point or disk is fixed in each of the three support directions; a tie is an endpoint where a maximizer changes.

Proof. By proposition 5.1, the side is the three-term support sum divided by $h = \sqrt{3}/2$. Parametrize unit normals by $n(\theta) = (\cos \theta, \sin \theta)$. For a disk of radius $\rho > 0$, its support value is ρ in every unit direction. On a polygon cell with fixed support vertices the sum is $g(\theta) = a \cos \theta + b \sin \theta > 0$, so $g'' = -g < 0$ and no interior minimum occurs. In the disk case a cell with q disk supports has sum $q\rho + \langle v, n(\theta) \rangle$, where v is the sum of the active point vectors rotated back by their respective multiples of 120° . If a point is active its projection exceeds $\rho \geq 0$, and the inner product is positive, again giving a negative second derivative. A disk-only cell has constant support sum 3ρ . Each application must still identify its exposed ties. $\square$

5.2.1. *Finite caliper criterion.* Let J denote counterclockwise rotation through $\pi/2$. For a nonempty finite set F define

$$(13) \quad W_F(N) := \sum_{k=0}^2 \max_{z \in F} \langle z, R^k N \rangle.$$

Theorem 5.3 (Finite caliper certificate). *If F has at least two distinct points, then F is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in F$ and $\varepsilon \in \{-1, 1\}$ such that*

$$(14) \quad W_F(\varepsilon J(q - p)) \leq \frac{\sqrt{3}}{2} \|p - q\|.$$

A singleton is contained in equilateral triangles of arbitrarily small positive side length.

Proof. Apply proposition 5.1 and lemma 5.2 to $\text{conv} F$. A minimizing normal is perpendicular to a hull edge $[p, q]$ and, after cyclic rotation, equals $\varepsilon J(q - p)/\|p - q\|$. Homogeneity gives (14); a nondegenerate segment follows by checking its two normals. $\square$

For a lower bound at a candidate normal N , it is enough to choose $p_0, p_1, p_2 \in \text{conv} F$ and use

$$W_F(N) \geq \sum_{j=0}^2 \langle p_j, R^j N \rangle.$$

Thus the maximizing point need not be identified in every direction. The candidate orientation list must still be exhaustive.

Proposition 5.4 (Disk-finite-set calipers). *Let $\rho > 0$, $P = \{p_1, \dots, p_m\} \subset \mathbb{R}^2$, and $K = \text{conv}(\mathbb{B}(O, \rho) \cup P)$. Use the rotation J defined above. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm \frac{J(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \neq p_k)$$

or

$$\frac{\rho p_i \pm \sqrt{\|p_i\|^2 - \rho^2} J p_i}{\|p_i\|^2} \quad (\|p_i\| \geq \rho).$$

Outside the disk-only alternative, a least enclosing equilateral triangle can be chosen so that one side contains two distinct points of P , or is tangent to the disk and contains one point of P .

Proof. Apply lemma 5.2. A point–point tie has normal perpendicular to $p_k - p_i$; solving $\langle p_i, n \rangle = \rho$ with $\|n\| = 1$ gives the displayed point–disk normals. Rotation through 120 degrees identifies the three supporting directions. A disk-only cell contributes 3ρ and is the only additional alternative. $\square$

How to use the finite list. For points, evaluate the three-support sum at the listed pair normals. For a disk and points, also test the point–disk ties. Nonexposed ties may be discarded, but their completeness must first be justified. The disk-only alternative contributes 3ρ only at an orientation where all three disk supports dominate every point; it is not an unconditional candidate value. Equivalently, adjoining one arbitrary reference normal to the complete tie list and evaluating the actual support sum at every listed normal gives a finite test, including the case with no ties. The final contact description in the proposition applies to its non-disk-only alternative.

One disk and one point. A useful explicit specialization, with $r > 0$ and $d = \|q\|$, is

$$\Lambda(\mathbb{B}(O, r) \cup \{q\}) = \begin{cases} 2\sqrt{3}r, & d \leq 2r, \\ \sqrt{3}r + \sqrt{d^2 - r^2}, & d > 2r. \end{cases}$$

Indeed, when $d \leq 2r$, orient the point projections as $d/2, d/2, -d$, so all three disk supports suffice. When $d > 2r$, the support-cell argument puts a minimum at a point–disk tie. If one projection equals r , the other two are $(-r \pm \sqrt{3(d^2 - r^2)})/2$; substitution in the support sum gives the second formula. A circle gives the same enclosure problem as its closed disk, since triangles are convex. This example explains the transition but does not replace the disk–finite-set proposition: several points can impose a joint obstruction that none imposes alone.

The other structural step. The fixed radial-forcing lemma and common-pair domination below turn local reach bounds into points missed by every open V triangle. The uniform common-pair corollary then supplies a symmetric inner hexagon and its inscribed disk. These are witness-construction results, not consequences of the finite caliper criterion. BC and D subsequently use fixed point sets with no disk; F uses nine fixed points and a disk-plus-three-point comparison set.

5.3. Local capacities and radial witnesses. The following capacities answer a local question: after boundary anchors have been prescribed, how far toward the center can a unit triangle reach? They are maxima over a family of triangles, not the actual reaches of the original triangle. A larger prescribed anchor shrinks the feasible family. The neighboring capacities are needed because a point on one radial segment may be covered by the V triangle at either adjacent vertex.

Normalize at V_0 and define the four-anchor hull

$$K(a, b, c) := \text{conv}\{V_0, X_5(1-a), X_0(b), (1-c)V_0\},$$

$$\mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

For

$$0 \leq a, b \leq 1, \quad a^2 + ab + b^2 \leq 1,$$

put

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

Write $f(a, b) := 1 - c_{\max}(a, b)$ for the radial deficit when $a + b \leq 1$. The *diameter envelope* is

$$(15) \quad M_0(x) = \frac{-x + \sqrt{4 - 3x^2}}{2} \quad (0 \leq x \leq 1).$$

Here M_0 without an argument denotes the radial midpoint, whereas $M_0(x)$ denotes this scalar function. For later rescuer estimates define

$$(16) \quad M_c^{\sup} := \frac{c + \sqrt{c^2 - 8c + 4}}{2} \quad (0 \leq c \leq 1/2).$$

The value at $c = 1/2$ is a continuous endpoint convention; no supercritical triangle has that radial demand. The bound associated with this envelope is proved in lemma A.9. The exact admissible set and radial envelope are proved in proposition A.7.

The neighboring-arm capacities are the variational quantities

$$C_+(a, b) := \max \left\{ c \in [0, 1] : \begin{array}{l} \exists T \text{ [} T \text{ is a closed unit equilateral triangle,} \\ \{V_0, X_5(1-a), X_0(b), (1-c)V_1\} \subseteq T \end{array} \right\},$$

$$C_-(a, b) := C_+(b, a), \quad 0 \leq a, b \leq 1, \quad a + b \leq 1.$$

Only a bound, not the full piecewise neighboring capacity, is needed below. The direct four-caliper proof is lemma D.1.

Lemma 5.5 (Common-pair domination). *If $p, q \geq 0$ and $p + q \leq 1$, then*

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

Proof. Put $m = \min(p, q)$. Convexity shortens the boundary anchors to (m, m) , so lemma D.1 gives $C_{\pm}(p, q) \leq 1 - m$. For the own-ray bound, use coordinates with origin V_0 and basis $V_5 - V_0, V_1 - V_0$. If $q = m$, the unit triangle $\text{conv}\{(-m, 0), (1 - m, 0), (1 - m, 1)\}$ contains the origin, $(p, 0)$, $(0, m)$, and the own-ray point $(1 - m, 1 - m)$, since $p \leq 1 - m$. Thus $c_{\max}(p, q) \geq 1 - m$; reflect if $p = m$. Increasing an anchor shrinks the feasible family, so capacities are antitone. $\square$

Lemma 5.6 (Strict own-ray bound with positive slack). *If $a, b > 0$ and $a + b < 1$, then*

$$c_{\max}(a, b) > 1 - \min(a, b).$$

Proof. Put $m = \min(a, b)$, $M = \max(a, b)$, and $\delta = 1 - a - b > 0$. By lemma 5.5, $c_{\max}(a, b) \geq 1 - m$. The exact local formula proposition A.7 has two branches. In the quartic branch, the defining polynomial $F(C, m) = C^4 - C^2 + mC - m^2$ satisfies

$$F(1 - m, m) = -m((1 - m)^3 + m^2) < 0.$$

In the triangular branch, the polynomial $H(C) = ((1 - \delta)^2 - 1)C^2 + MC - M^2$ satisfies

$$H(1 - m) = -\delta((1 - m)(1 - 2m) + m(2 - m)\delta) < 0.$$

Thus equality with $1 - m$ is impossible on either selected capacity branch. The non-strict bound proves the strict inequality. This argument does not select a geometric component from a polynomial sign; it excludes equality for the capacity already specified by the exact local theorem. $\square$

Definition 5.7 (Fixed total radial endpoint). Parametrize r_i by $Z_i(c) = (1 - c)V_i$, measured from V_i toward O . Let $u_{j \rightarrow i}$ be the largest parameter in the actual positive trace of T_j on r_i for $j = i - 1, i + 1$, with value zero when no such positive trace exists. Put

$$\gamma_i := \max\{C_i, u_{i-1 \rightarrow i}, u_{i+1 \rightarrow i}\}, \quad d_i := 1 - \gamma_i, \quad \hat{P}_i := d_i V_i.$$

Thus $\hat{P}_i$ is the innermost endpoint reached by any local V triangle on r_i , rather than just the endpoint of the own V triangle. This distinction is essential when a nonsupercritical V triangle supports an adjacent radial segment.

Lemma 5.8 (Total-endpoint radial forcing). *One has $0 < d_i < 1$, and every sV_i with $0 \leq s \leq d_i$ is missed by all open V triangles. Under skeleton coverage these points belong to U_C .*

Proof. No closed V triangle contains O : moving its interior distinguished vertex slightly away from O would contradict diameter one. Each own trace has positive length, so $0 < \gamma_i < 1$ and $0 < d_i < 1$. For $0 < s \leq d_i$, containment of sV_i in a local open role would extend its trace to an inward parameter strictly above $1 - s \geq \gamma_i$. This contradicts maximality, including the case of an absent positive neighboring trace. Nonlocal roles are excluded by

$$\|sV_i - V_{i \pm 2}\|^2 = 1 + s + s^2 > 1, \quad \|sV_i - V_{i+3}\| = 1 + s > 1.$$

The case $s = 0$ is the already-excluded center. Skeleton coverage supplies U_C . $\square$

Corollary 5.9 (Clipped radial forcing). *For $j = i - 1, i, i + 1$, let $0 \leq a_j \leq A_j$, $0 \leq b_j \leq B_j$, and $a_j + b_j \leq 1$. Define*

$$s_i := \min\{f(a_i, b_i), a_{i-1}, b_{i-1}, a_{i+1}, b_{i+1}\}.$$

Every sV_i with $0 \leq s \leq s_i$ is missed by all open V triangles. For actual reaches on a gap-free nonsupercritical path, this simplifies to

$$(17) \quad s_i = \min\{f(A_i, B_i), A_{i-1}, B_{i+1}\}.$$

Proof. The own endpoint is at most $1 - f(a_i, b_i)$. By common-pair domination, each positive neighboring endpoint is at most $1 - \min(a_j, b_j)$; an absent trace contributes zero. Hence $\gamma_i \leq 1 - s_i$ and lemma 5.8 applies. On a gap-free nonsupercritical path, A increases and B decreases by the boundary deficit identities. Together with $f(A_i, B_i) \leq \min(A_i, B_i)$, these inequalities remove B_{i-1} and A_{i+1} from the minimum. $\square$

The actual radius d_i comes from maximal traces; $s_i \leq d_i$ is a certified lower bound. Neither may be replaced by the own-ray radius without accounting for neighboring suppliers.

Corollary 5.10 (Uniform common-pair radial forcing). *Suppose $p, q \geq 0$, $p + q \leq 1$, $p \leq A_i$, $q \leq B_i$ for every i , and $0 < c_* := c_{\max}(p, q) < 1$. Then $(1 - c_*)V_i$ is missed by all open V triangles, without a restriction on their normalized types. Under skeleton coverage,*

$$(1 - c_*)V_i \in U_C \quad (0 \leq i \leq 5), \quad \mathbb{B}(O, h(1 - c_*)) \subset U_C.$$

Proof. Use the selected pair (p, q) at every vertex in corollary 5.9. Since $f(p, q) \leq \min(p, q)$, the clipped radius equals $1 - c_*$. Convexity of U_C supplies the inner hexagon and its displayed inscribed disk. $\square$

5.4. Fixed witnesses and candidate enclosures. Each witness set is fixed from the original V triangles before an arbitrary enclosing candidate is chosen. The finite calipers vary their support orientation, but the witness points do not. Gap endpoints are supplied by lemma 2.10; radial witnesses use the total endpoints, including all permitted neighboring traces.

Replacement is a separate skeleton-preserving operation, not a fourth witness construction. BC handles the center-aligned case. For an off-center supercritical role, its midpoint supplier leads to D, replacement, or the Vd2 perimeter obstruction.

5.5. A selected gap forces five points (BC). Normalize the center midpoint to M_0 and choose any actual gap, reflected to J_0 . The other incident edge $e_{5,0}$ may or may not contain a gap. Only the four middle edges $e_{1,2}, \dots, e_{4,5}$ must be gap-free. The following transfer supplies the tail bound needed by the path argument.

Lemma 5.11 (Shared-gap-anchor boundary transfer). *If the original open roles cover ∂H and J_0 is an actual gap, then, independently of the gap status of $e_{5,0}$,*

$$B_5 > 1 - M_0(B_0) > B_0/2.$$

Here $M_0(z)$ is the scalar diameter envelope.

Proof. Put $G = X_0(B_0)$ and $L = X_5(B_5)$. Actual gap forcing gives $G \in U_C$, while maximality gives $G \in T_0$ and $L \notin U_5$. Boundary locality and the original perimeter cover give $L \in U_0 \cup U_C$. Thus one unit triangle contains L openly and G in its closure, so $\|G - L\| < 1$. Indeed, at equality one could move L slightly away from G inside that open triangle and exceed its diameter. The two incident-edge directions at V_0 meet at 120 degrees, hence

$$B_0^2 + B_0(1 - B_5) + (1 - B_5)^2 < 1.$$

Apply lemma A.12; its last inequality is strict because $B_0 > 0$. The proof includes singleton gaps and does not require L to be a witness in the final finite set. $\square$

In the pure enclosure statement below the tail bound $B_5 \geq B_0/2$ is explicit. In the covering application, the preceding lemma establishes it for the original roles before an arbitrary enclosing candidate is chosen. We do not assume that the candidate also covers the discarded gap.

Define the two nonuniform comparison pairs and radii by

$$x = B_0, \quad y = 1 - A_1, \quad z = B_3.$$

$$r = 1 - c_{\max}(1 - y, z), \quad t = 1 - c_{\max}(1 - z, x/2).$$

Put $\hat{t} = \min\{t, A_3\}$ and fix

$$K_{BC} := \{M_0, X_0(x), X_0(y), rV_2, \hat{t}V_4\}.$$

There are at most five points, including singleton gaps. The midpoint is a structural anchor; the other points are missed by every open V role. The cap by A_3 controls the remaining neighboring supplier on r_4 ; it is not silently dropped.

The five-point selected-gap construction. Set $x = B_0$, $y = 1 - A_1$, and $z = B_3$. The two boundary pairs $(1 - y, z)$ and $(1 - z, x/2)$ are different; they preserve the shared boundary parameter z instead of replacing all path data by a uniform minimum. Their capacity deficits give r and t . The five witnesses are $M_0, X_0(x), X_0(y), rV_2, \min(t, A_3)V_4$. There is no point on r_3 , although B_3 remains

part of the construction. The cap by A_3 handles a neighboring supplier. When it is active, a pair of witnesses already has distance greater than one; otherwise the capacity inequality verifies the single decisive caliper of the five-point threshold criterion. Singleton gaps merely identify $X_0(x)$ and $X_0(y)$.

Theorem 5.12 (Unified five-point selected-gap enclosure). *Suppose J_0 is an actual gap, $A_i + B_i \leq 1$ for $1 \leq i \leq 5$, and*

$$B_i + A_{i+1} > 1 \quad (1 \leq i \leq 4), \quad B_5 \geq B_0/2.$$

Then $\Lambda(K_{BC}) > 1$. Under skeleton coverage with distinguished C midpoint M_0 , the fixed set K_{BC} lies in U_C . There is no condition on the gap status of $e_{5,0}$ or supercriticality of T_0 , and no type restriction on the nonsupercritical path.

Proof. The boundary-deficit identities give increasing $A_1, \dots, A_5$ and decreasing $B_1, \dots, B_5$. Thus $0 < x \leq y < 1$ and $x/2 \leq z \leq y$. Put $\rho_i = f(A_i, B_i)$. By corollary 5.9, the safe radii are $\min\{\rho_i, A_{i-1}, B_{i+1}\}$ for $i = 2, 3, 4$. The bounds $(A_2, B_2) \geq (1 - y, z)$ imply $r \leq \rho_2$, $r \leq A_1$, and $r \leq B_3$. Hence rV_2 is safe. Similarly $(A_4, B_4) \geq (1 - z, x/2)$ gives $t \leq \rho_4$, and $t \leq x/2 \leq B_5$; clipping by A_3 makes tV_4 safe. The gap endpoints follow from lemma 2.10, and the midpoint belongs to U_C by hypothesis. All witnesses are fixed before enclosure is tested.

If $A_3 \geq t$, apply the capacity instance of lemma D.5. Otherwise $A_3 < t$ and $A_3 \geq A_1 = 1 - y$, so

$$\|X_0(y) - A_3V_4\|^2 \geq \|X_0(y) - (1 - y)V_4\|^2 = 1 + (1 - y)(2 - y) > 1.$$

The diameter obstruction gives $\Lambda(K_{BC}) > 1$ in this case also. No signed center coordinates or CE1 return are used in this enclosure argument. $\square$

Theorem 5.13 (Center-aligned path obstruction). *Suppose a skeleton cover has at least one actual gap and its C triangle's unique midpoint is M_k . If every T_i with $i \neq k$ is nonsupercritical, the configuration is impossible.*

Proof. Normalize $k = 0$. The signed normal form confines possible gap edges to $e_{5,0}, e_{0,1}$. Select either actual gap and reflect it to J_0 . The four middle edges are gap-free, and lemma 5.11 gives $B_5 > B_0/2$. Apply theorem 5.12. The midpoint belongs openly to U_C , and all other witnesses are center-forced. Compact-open containment contradicts $\Lambda(K_{BC}) \geq 1$ in either gap rank. $\square$

Theorem 5.14 (Every skeleton cover has a supercritical V triangle). *A skeleton cover by the seven original open roles satisfies $N_+ \geq 1$.*

Proof. If $N_+ = 0$ and there are no gaps, strict overlaps give $6 < \sum_i (A_i + B_i) \leq 6$. If a gap is present, all six roles are nonsupercritical, so theorem 5.13 applies. This proof uses only BC; replacement will invoke N0, not conversely. $\square$

Lemma 5.15 (A unique midpoint supplier). *In a nonzero-gap skeleton cover, the preceding N0 theorem and the skeleton budget leave only $N_+ = 1$, $N_{sp} \leq 1$. Write M_k for the C midpoint and σ for the supercritical index. If $\sigma \neq k$, there is a unique positive-support role T_τ , where*

$$\tau \in \{\sigma - 1, \sigma + 1\}, \quad M_\sigma \in U_\tau.$$

If T_τ is T3-like, then $\tau = k$. If $\tau \neq k$, the edge shared by T_τ, T_σ is center-free, in both CE1 and CE2.

Proof. N_0 excludes $N_+ = 0$. Two supercritical roles would require a distinct positive-support rescuer by lemma 3.6, and theorem 3.7 would exclude them. The same count bound gives $N_{\text{sp}} \leq 1$.

If $\sigma \neq k$, neither the C triangle nor T_σ contains M_σ . Diameter excludes nonlocal roles; a Vd0 role has no positive adjacent trace and cannot contain that point openly. Hence its unique supplier is an adjacent positive-support role. A T3-like supplier also misses its own midpoint by lemma 2.11. Unless $\tau = k$, neither the center nor the other roles could cover this second midpoint: all other roles are Vd0. Finally, when $k \notin \{\tau, \sigma\}$, the shared edge is not incident to V_k , whereas every open center boundary trace is incident to V_k by the signed normal form. $\square$

5.6. A supported midpoint forces four points (D). Let $Y(t) = (1-t)V_0 + tV_5 = X_5(1-t)$. The geometric lemma below is stated independently of any V type or covering configuration.

Theorem 5.16 (Four-point rescuer geometry). *Suppose $a \geq 0$, $\varepsilon > 0$, $\beta \geq 0$, and*

$$s = a + \varepsilon, \quad \beta \leq \frac{\varepsilon}{s}.$$

Then

$$\Lambda\{O, \varepsilon V_1, Y(a), Y(1-\beta)\} \geq 1.$$

Neither $a \leq \varepsilon$ nor $s \leq 1$ is required by this geometric statement.

Proof. If $a = 0$, the set contains O, V_0 , at distance one. If $s \geq 1$, then

$$\|Y(a) - \varepsilon V_1\|^2 = 1 - s + s^2 \geq 1.$$

These are diameter obstructions. Otherwise put $v = a/s$, so $0 < s, v < 1$. The ratio gives $1 - \beta \geq v$, while $a = sv \leq v$. Hence $Y(v)$ lies on $[Y(a), Y(1-\beta)]$. By convexity it suffices to enclose the quadrilateral, in cyclic order,

$$O, \quad Q = Y(v), \quad A = Y(sv), \quad P = s(1-v)V_1.$$

With $h = \sqrt{3}/2$, its outward edge normals may be chosen as

$$(-hv, v/2 - 1), \quad v(1-s)(h, -1/2), \quad (hs, 1 - s/2), \quad s(1-v)(-h, 1/2).$$

The maximizing triples in each direction and its two 120-degree rotations are (O, A, P) , (A, P, O) , (A, O, Q) , and (O, Q, A) , respectively. The resulting caliper sides are

$$\frac{1}{\sqrt{1-v+v^2}}, \quad 1+s(1-v), \quad \frac{1}{\sqrt{1-s+s^2}}, \quad 1+v(1-s).$$

All exceed one for $0 < s, v < 1$. The finite-caliper theorem theorem 5.3 completes the nondegenerate case. $\square$

Why four D points suffice. The two gap endpoints and the forced supported-ray endpoint are fixed from the original triangles. The geometric condition is only $\beta \leq \varepsilon/(a + \varepsilon)$. If $s = a + \varepsilon \geq 1$, a pair has squared distance $1 - s + s^2 \geq 1$. For $0 < s < 1$, put $v = a/s$: convexity allows the far endpoint to move inward to $Y(v)$, and the four resulting hull-edge calipers all have side greater than one. The original supplier charts still establish forcing and the ratio; no candidate center normalization is used in the enclosure calculation.

Lemma 5.17 (Scalar rescuer ratio test). *For $0 \leq x, c \leq 1/2$ and $M = (c + \sqrt{c^2 - 8c + 4})/2$,*

$$x \leq 1 - M \iff x^2 + (c - 2)x + c \geq 0.$$

Proof. The smaller root is $1 - M$, while the other root is at least one. The quadratic is nonnegative on $[0, 1/2]$ exactly before its smaller root. $\square$

For the covering application, let T_0 supply M_1 , with supported interval $[c, u]$ on r_1 , measured from V_1 toward O , and put $\varepsilon = 1 - u$ and $P_T = \varepsilon V_1$. All interval endpoints here belong to the original triangle, not to its translated majorant. The local T3-like and Vd1 calculations establish

$$A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M_c^{\sup}.$$

Together with $C_1 \geq c$, the boundary path, and the forced endpoint, lemma D.6 gives the single fixed witness

$$K_D = \{O, \varepsilon V_1, Y(A_0), Y(1 - B_5)\} \subset U_C, \quad \Lambda(K_D) \geq 1.$$

The two local verifications are lemmas D.7 and D.8; the enclosure step does not distinguish one from two gaps.

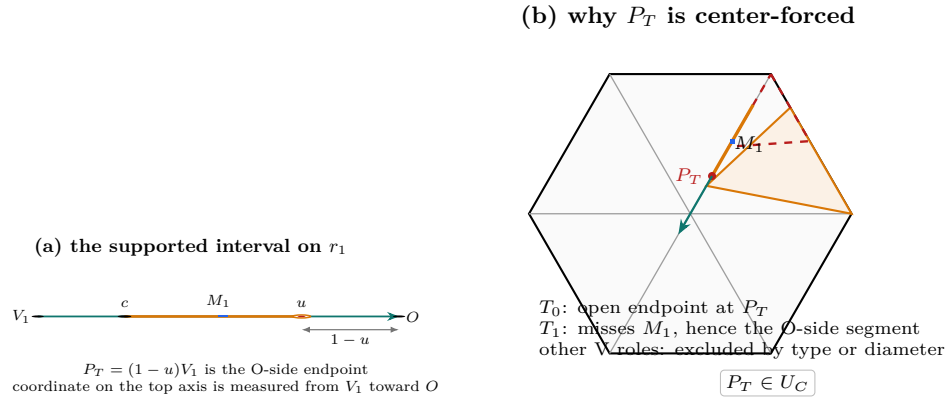


FIGURE 4. Construction of the T3-like rescuer endpoint. The interval $[c, u]$ is parametrized from V_1 toward O , so its O-side endpoint is $P_T = (1 - u)V_1$. The T3-like open role omits its endpoint, the adjacent supercritical role contains V_1 but misses M_1 and therefore misses the O-side segment by convexity, and all remaining V roles are excluded. Thus P_T is forced into the C triangle.

A change of configuration, not a new candidate. The away-supplier case uses a different operation from finite enclosure. Under the stated two-chart hypotheses, two V triangles are replaced while the distinguished roles and coverage of the skeleton are retained. The result in this unique-supercritical case has no supercritical V triangle, contradicting theorem 5.14. This explains why that theorem was proved for *skeleton* covers before replacement was introduced.

Neither coverage of all of H nor the original number of gaps is claimed to survive. The contradiction uses only the preserved skeleton cover. Likewise, the earlier translated closed-trace majorant is not itself a valid replacement open triangle: its

distinguished vertex lies on its boundary. These three objects—a majorant, an enclosing candidate, and a replacement arrangement—have different uses.

5.7. The zero-gap nine-point obstruction. The remaining zero-gap case has one supercritical V triangle. There is no boundary gap to use as a witness, so the obstruction must be built inside H . The construction and its verification have three stages.

Force the points. Strict boundary handoffs give common lower bounds for the boundary reaches. These force six radial points into the C triangle. Three further points are chosen on the exact frontier of the feasible union of the supercritical triangle; separate diameter and handoff arguments exclude every other V triangle. Exclusion from the supercritical triangle alone would not be enough. A frontier point may belong to a closed feasible triangle while being missed by its open interior.

Simplify the set to be enclosed. The convex hull of the six radial points contains a disk. Replacing those six points by this disk gives a smaller comparison set. In the nontrivial range, the two outer frontier points are then moved inward by a single Newton step at the fixed line parameter $k = 1/2$. Convexity keeps these comparison points inside any candidate containing the original witnesses. These are exact inner comparisons, not numerical approximations to a covering configuration.

Bound the enclosing triangle. The inner comparison reduces the enclosure problem to four supporting contacts: two contacts along consecutive-point lines and two disk-tangent contacts. The line estimates are analytic. The remaining paired tangent inequalities are reduced to the exact polynomial certificate. At the end, the comparison set requires side at least one, contradicting its containment in the open C triangle. The calculations below justify each of these stages; the certificate enters only at the last one.

Active tangent calculation. The two outer comparison points use the fixed-start formulas, not the older junction-start formulas. The exact coefficient comparisons and radius envelope are specified in theorem F.6. The actual disk is unchanged; the smaller scalar radius is used only for initially checking the tangent residuals.

Assume

$$N_{\text{gap}} = 0, \quad N_+ = 1.$$

Rotate the unique supercritical role to index 4. The exact-one handoff calculation lemma E.2 gives

$$(18) \quad \xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

Define

$$(19) \quad a := 1 - \xi_3, \quad b := \xi_4, \quad p := 1 - b, \quad q := 1 - a.$$

Then

$$(20) \quad \begin{aligned} 0 < a, b < 1, \quad a &= a_4 < A_4, \quad b = b_4 < B_4, \\ a + b &> 1, \quad a^2 + ab + b^2 < 1, \\ a_i &\geq p, \quad b_i \geq q \quad (0 \leq i \leq 5). \end{aligned}$$

Define the two boundary anchors

$$P_A := V_4 + a(V_3 - V_4), \quad P_B := V_4 + b(V_5 - V_4),$$

and the feasible union

$$\mathcal{R}_{AB}(a, b) := \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Put

$$c_* := c_{\max}(p, q), \quad r_* := h(1 - c_*), \quad P_i^{\text{rad}} := (1 - c_*)V_i \quad (0 \leq i \leq 5).$$

The bounds in lemma E.3 give $0 < c_* < 1$. Applying corollary 5.10 to the common pair gives

$$(21) \quad \mathbb{B}(O, r_*) \subseteq \text{conv}\{P_0^{\text{rad}}, \dots, P_5^{\text{rad}}\} \subset U_C.$$

The two circles adjacent to the supercritical frontier are

$$\mathcal{C}_- := \partial\mathbb{B}(X_2(b), 1), \quad \mathcal{C}_+ := \partial\mathbb{B}(X_5(1 - a), 1).$$

Let ℓ_-, ℓ_+ be the two straight pieces of the interior frontier of $\mathcal{R}_{AB}(a, b)$, in the order of theorem E.1. Define

$$\{Q_0\} = \ell_- \cap \ell_+, \quad \{Q_-\} = \ell_- \cap \mathcal{C}_-, \quad \{Q_+\} = \ell_+ \cap \mathcal{C}_+.$$

The coordinate formulas and selected first roots are given in Appendix E, equations (69) and (74). Lemma E.5 proves that these intersections are singletons on the indicated pieces. The nine-point set is

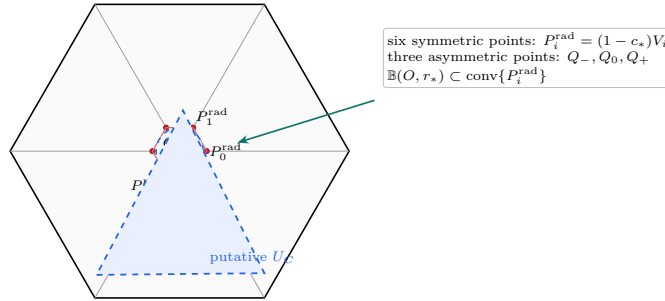
$$K_F := \{P_i^{\text{rad}} : 0 \leq i \leq 5\} \cup \{Q_-, Q_0, Q_+\}.$$

Their exact construction and exclusions in theorem E.1 and lemmas E.5 and E.6 use contained vertices and actual handoffs, not Vd0 locality, and give

$$(22) \quad Q_-, Q_0, Q_+ \in U_C.$$

For the enclosure calculation use the compact comparison set

$$(23) \quad K_{\text{wit}}(a, b) := \mathbb{B}(O, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$



$K_{\text{wit}} = \mathbb{B}(O, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C$ under a hypothetical cover

FIGURE 5. The type-independent zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB -frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

The witness set and the comparison set are different. The original finite witness is K_F , with at most nine points. The set K_{wit} contains a disk in place of the six radial points and satisfies

$$K_{\text{wit}} \subseteq \text{conv}(K_F), \quad \Lambda(K_F) = \Lambda(\text{conv}(K_F)) \geq \Lambda(K_{\text{wit}}).$$

Thus proving the lower bound for the disk comparison proves it for the nine-point witness. The disk need not be missed by every V triangle: its containment in U_C follows from convexity and the six forced radial points. The later Newton points play the same comparison role; they do not replace the definitions of the frontier points Q_-, Q_0, Q_+ .

Active tangent calculation. The two outer comparison points use the fixed-start formulas, not the older junction-start formulas. The exact coefficient comparisons and radius envelope are specified in theorem F.6. The actual disk is unchanged; the smaller scalar radius is used only for initially checking the tangent residuals.

Why four contacts suffice in the hard range. The additional geometric step is not another general enclosure formula. The fixed-start Newton inner reduction moves the two outer frontier witnesses inward, preserving the candidate's containment by convexity. Their convex hull with the disk has an exposed three-point chain; its two successive point-point lines and two outer disk tangencies supply the four contacts. The support-cell lemma then makes that contact list exhaustive. The inner inclusion, contact geometry, and tangent inequalities must still be proved; their calculations are in lemmas E.7 and E.11 and proposition E.8. Four contacts are not four witness points, and the exact tangent certificate is not replaced by the finite-candidate theorem.

Theorem 5.18 (Witness enclosure). *For $0 < a, b < 1$, $a+b > 1$, and $a^2+ab+b^2 < 1$, the comparison set $K_{\text{wit}}(a, b)$ satisfies*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Proof. The disk closes the range $c_* \leq 2/3$. For $c_* > 2/3$, the Newton inner points, constructed from the fixed parameter $k = 1/2$, reduce the enclosure to four supporting contacts: the lines through two consecutive points and the two exposed disk tangencies. The line bounds are analytic; the paired tangent residuals use explicit low-degree comparisons with exact coefficient bounds and simultaneous radius transfer. The complete calculation is lemma E.11. $\square$

Theorem 5.19 (Zero-gap obstruction). *There is no hypothetical cover with*

$$N_{\text{gap}} = 0, \quad N_+ = 1.$$

Proof. The construction gives the compact containment $K_{\text{wit}}(a, b) \subset U_C$, while theorem 5.18 gives $\Lambda(K_{\text{wit}}(a, b)) \geq 1$. This contradicts lemma 2.1. $\square$

5.8. Summary of the fixed witnesses.

6. COMPLETION OF THE PROOF

Proof of Theorem 1.1. Suppose seven open unit equilateral triangles cover H . By theorem 5.14, $N_+ \geq 1$. With no gaps, theorem 5.19 excludes $N_+ = 1$, and proposition 4.3 excludes $N_+ \geq 2$.

With a gap, lemma 5.15 leaves $N_+ = 1, N_{\text{sp}} \leq 1$. Let M_k be the C midpoint and σ the supercritical index. If $\sigma = k$, apply theorem 5.13. Otherwise the supercritical

midpoint has a unique adjacent supplier. A T3-like supplier is center-based. In either center-based T3-like or Vd1 placement, the corresponding actual-endpoint calculation lemmas D.7 and D.8 and the common adapter lemma D.6 give D. A Vd2 supplier is excluded by proposition 3.9. An away Vd1 supplier has a center-free shared edge; the two-chart replacement proposition D.11 yields a skeleton cover with $N'_+ = 0$, contradicting N0. This uses BC to prove N0 before replacement, not replacement to prove N0. Thus no cover exists. The scaling equivalence proposition 2.2 gives corollary 1.2. $\square$

APPENDIX A. STRUCTURAL, SHARED-LOCAL, AND SIGNED-CENTER
OPTIMIZATION

This appendix solves the coordinate problems used by the geometric interfaces in Section 2. The shared equilateral-enclosure calculus is proved in the main text, Subsection 5.2. Each calculation here is attached to the body statement it supplies.

A.1. Center and V triangle incidence feasibility.

Lemma A.1 (Separation of nonadjacent boundary edges). *Relative-interior points on two nonadjacent edges of H are more than unit distance apart.*

Proof. Up to symmetry take

$$x = (1 - t)V_0 + tV_1, \quad 0 < t < 1.$$

For $y = (1 - s)V_2 + sV_3$, $0 < s < 1$, direct expansion gives

$$\|x - y\|^2 - 1 = (1 - t)(2 - t) + s(s + t) > 0.$$

For the other orbit, $y = (1 - s)V_3 + sV_4$, put $u = t + s$ and obtain

$$\|x - y\|^2 - 1 = (u - 1)^2 + 2 > 0.$$

These are the two nonadjacent-edge configurations. Since the diameter of a unit equilateral triangle is 1, the result supplies Proposition 2.4. $\square$

Lemma A.2 (Shared corner chart and wedge reduction). *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \quad W = \{(x, y) : x \geq 0, y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0 , then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y) -coordinate area.

Proof. The two basis vectors have length one and angle 120° , so

$$\|(x, y)\|^2 = x^2 + y^2 - xy, \quad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \leq x, y \leq 2, |x - y| \leq 1\} \subset W.$$

Conversely, if $(x, y) \in T \cap W$, then the diameter-one condition gives $x^2 + y^2 - xy \leq 1$. Hence

$$|x - y|^2 \leq x^2 + y^2 - xy \leq 1, \quad \frac{3}{4}x^2 \leq x^2 + y^2 - xy,$$

and the same estimate holds for y . Thus $|x - y| \leq 1$ and $0 \leq x, y \leq 2/\sqrt{3} < 2$, proving $(x, y) \in H$. The determinant identity proves the area assertion. $\square$

Lemma A.3 (Local wedge calculation). *The incidence conclusions of Lemma 2.5 hold.*

Proof. Use lemma A.2 at V_0 . In the unit ball,

$$x^2 + y^2 - xy = (y - x/2)^2 + 3x^2/4 = (x - y)^2 + xy.$$

The adjacent radial segments are

$$r_1 = \{(1, y) : 0 \leq y \leq 1\}, \quad r_5 = \{(x, 1) : 0 \leq x \leq 1\}.$$

Suppose $T_0 \cap r_1$ has positive length and let m be the largest x -coordinate of a vertex of T_0 . If $m > 1$, a maximizing vertex (m, y) lies in the unit ball and has $0 < y < 1$, so it lies in H . If $m = 1$, positive-length contact with the support line $x = 1$ forces a side there, and both endpoints satisfy $1 + y^2 - y \leq 1$ and lie in H . The argument for r_5 is symmetric. If both arms are met but only one vertex lies in H , that vertex would have $x > 1$ and $y > 1$, contradicting $(x - y)^2 + xy > 1$.

Finally, if all three vertices lay in convex H , then $T_0 \subset H$, contrary to the extreme point V_0 lying in $\text{int}(T_0)$. Hence $o(T_0) \geq 1$, completing the calculation. $\square$

A.2. Shared orientation charts and exact-trace normalization. Set

$$D_t := 1 - t + t^2, \quad z := \sqrt{D_t}, \quad 0 \leq t \leq 1.$$

In the shared corner chart, physical rotation through 60° is $(x, y) \mapsto (x - y, x)$. After relabelling the vertices, every orientation belongs to one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

They cover successive 60° direction sectors, including their common endpoints. The following half-plane descriptions allow closed containment of the corner; strict interior containment is recorded separately. Here U, V are affine slack functions, not the open triangles U_i . In Type I write

$$(24) \quad U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\},$$

and in Type II write

$$(25) \quad U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

In both charts

$$(26) \quad V_0 \in \text{int}(T) \iff \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z.$$

The Type-II vertex in the wedge is

$$(27) \quad Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right).$$

Lemma A.4 (Raw $(3, 0)$ trace-normalization calculation). *The translation asserted in Proposition 2.6 exists and preserves both the open and closed traces in H .*

Proof. Normalize to $i = 0$. A Type-II triangle cannot have $o = 3$, because the point Q in (27) lies in the wedge; together with the diameter-one condition, Lemma 2.5 puts it in H . Hence T has Type-I form. Its vertices are

$$(28) \quad \begin{aligned} Q_0 &= \left(-\frac{(1 - t)\alpha + \beta}{D_t}, -\frac{\alpha + t\beta}{D_t} \right), \\ Q_1 &= \left(\frac{z - (1 - t)\alpha - \beta}{D_t}, \frac{tz - \alpha - t\beta}{D_t} \right), \\ Q_2 &= \left(\frac{(1 - t)z - (1 - t)\alpha - \beta}{D_t}, \frac{z - \alpha - t\beta}{D_t} \right). \end{aligned}$$

Thus $o = 3$ is equivalent to

$$(29) \quad \alpha + t\beta > tz, \quad (1 - t)\alpha + \beta > (1 - t)z,$$

which forces $0 < t < 1$. Put $s = \alpha + \beta$ and $L = z - s > 0$. Then

$$(30) \quad \alpha > \frac{tL}{1-t}, \quad \beta > \frac{(1-t)L}{t}.$$

For $(x, y) \in W$ satisfying

$$(31) \quad (1-t)x + ty \leq L,$$

the minimum of U is attained at $(L/(1-t), 0)$ and that of V at $(0, L/t)$. Hence

$$(32) \quad T \cap W = \{(x, y) \in W : (1-t)x + ty \leq L\}.$$

Keep t, s fixed and set

$$(33) \quad \hat{\alpha} := \frac{tL}{1-t}, \quad \hat{\beta} := s - \hat{\alpha}.$$

Then $0 < \hat{\alpha} < \alpha$, $\hat{\beta} > \beta > 0$, and $\hat{\alpha} + \hat{\beta} = s < z$. Changing these affine constants is a translation and preserves $V_0 \in \text{int}(\hat{T})$. Its third inequality remains (31); $\hat{U} = 0$ only at the endpoint $(L/(1-t), 0)$ of that common boundary and $\hat{V} > 0$. Therefore

$$\hat{T} \cap W = T \cap W,$$

and their interiors have the same trace on W . The local wedge calculation identifies these with the closed and open traces in H .

Equation (33) places Q_1 on the boundary of H ; Q_0 remains outside and the second strict inequality in (30) keeps Q_2 outside. Hence $(o(\hat{T}), n(\hat{T})) = (2, 0)$. Since the three reach segments lie in H , their open and closed traces, and therefore A_i, B_i, C_i and $A_i + B_i > 1$, are unchanged. $\square$

A.3. T3-like trace-majorant feasibility. The calculation uses the local wedge and the two orientation charts just introduced. Its output is a closed majorant, not a replacement open triangle.

Lemma A.5 (T3-like translation calculation). *The closed-trace majorant in Proposition 2.8 exists. In the calculation chart, every original T3-like triangle also satisfies $A_i + B_i \leq 1$ and $M_i \notin T_i$.*

Proof. Use the wedge chart of lemma A.3 and the orientation forms (24)–(25) in Subsection A.2. In Type I, direct inversion shows that when exactly two vertices lie outside W , the sole wedge vertex has both coordinates strictly below 1: the relevant bounds are $(1-t)/z$ and t/z , with strictness from $\alpha > 0$ or $\beta > 0$. The support argument in lemma A.3 then excludes positive-length contact with either adjacent radial line. Thus a T3-like triangle has Type II.

Reflect in $x = y$ if necessary so that $T_0 \cap r_1$ has positive length. The endpoint cases $t = 0, 1$ give no such interval, hence $0 < t < 1$. The exact axis reaches are

$$A_0 = \beta, \quad B_0 = z - \alpha - \beta,$$

and the support condition is

$$Q_x > 1, \quad Q_y \leq 1,$$

where Q is (27). In particular

$$A_0 + B_0 = z - \alpha < z \leq 1.$$

The same support inequality gives

$$\beta < \frac{z - D_t}{t} < \frac{1 - t}{2},$$

where the second comparison is equivalent to $(1 - z)^2 > 0$. At $M_0 = (1/2, 1/2)$ the affine coordinate V equals $\beta - (1 - t)/2 < 0$, so $M_0 \notin T_0$.

Define $\widehat{T}$ by replacing α with 0 and retaining t, β . This changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so V_0 lies in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1 - t)x + ty \geq 0, \quad \widehat{V} = V, \quad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old x -axis reach is B_0 and the new reach is $B_0 + \alpha$, so the inclusion is strict in H .

The two outside vertices remain outside, while V_0 and the new wedge vertex lie in H . Moreover $\widehat{Q}_x > Q_x > 1$, and $Q_x > 1$ gives

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1 - t)\alpha > 0.$$

Hence $\widehat{Q}_y < 1$ and $(o(\widehat{T}), n(\widehat{T})) = (2, 1)$. At every nonzero point of W , $\widehat{U} > 0$; the other two displayed slack relations show that old interior points other than V_0 remain interior. Reflection proves the other supported-arm case. $\square$

A.4. Strict-handoff feasibility.

Lemma A.6 (Cyclic strict-handoff selection). *Under the hypotheses of Proposition 2.12, all six overlap intervals are nonempty, and the selectors can preserve the stated one- or two-supercritical-index alternatives.*

Proof. Since V_i is interior to a diameter-one triangle, $0 < A_i < 1$ and $0 < B_i < 1$. No nonincident V triangle contains a relative-interior point of $e_{i,i+1}$. Thus the relatively open incident traces cover the edge and overlap, so

$$(1 - A_{i+1}, B_i) \neq \emptyset.$$

Choosing ξ_i in this interval places both selected anchors in U_i . Their squared distance is $a_i^2 + a_i b_i + b_i^2$, which is strictly below 1 because both are interior points of the same open unit triangle.

Put $\ell_i := 1 - A_{i+1}$ and $u_i := B_i$. Then

$$(34) \quad A_i + B_i > 1 \iff \ell_{i-1} < u_i, \quad a_i + b_i > 1 \iff \xi_{i-1} < \xi_i.$$

If i is actually nonsupercritical, then

$$\xi_i < u_i \leq \ell_{i-1} < \xi_{i-1},$$

so no selection makes it supercritical. Also

$$(35) \quad \sum_{i=0}^5 (a_i + b_i) = \sum_{i=0}^5 (1 - \xi_{i-1} + \xi_i) = 6.$$

If p is the sole actual supercritical index, the other five selected sums are below 1, and (35) forces the p th sum above 1.

For two nonadjacent supercritical indices p, q , choose

$$\ell_{r-1} < z_r < u_r \quad (r \in \{p, q\}),$$

and then

$$\xi_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \quad \xi_r \in (\max\{\ell_r, z_r\}, u_r).$$

The pairs are disjoint and create ascents at p, q . Any three indices on a six-cycle contain two nonadjacent indices. If the only two supercritical indices are adjacent, say $p, p+1$, the remaining inequalities give

$$\ell_{p-1} < \ell_{p+1}, \quad \max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose ξ_p between the latter bounds and choose $\xi_{p-1} < \xi_p < \xi_{p+1}$ in their overlap intervals. By (34), both selected indices remain supercritical. $\square$

The enclosure gauge, support-cell rotation, and finite calipers are stated and proved in Subsection 5.2 of the finite-enclosure chapter. The local optimization below applies those general results. For these reminders, write

$$W_K(n) := \sum_{j=0}^2 h_K(R^j n) \quad (\|n\| = 1).$$

For reference, the gauge formulas used here are

$$\Lambda(K) := \frac{2}{\sqrt{3}} \min_{\|n\|=1} W_K(n),$$

$$K \text{ is contained in a unit C triangle} \iff \min_{\|n\|=1} W_K(n) \leq \frac{\sqrt{3}}{2}.$$

For the disk-finite-set applications, use the same inverse scaling:

$$\Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} W_K(n), \quad K = \mathbb{B}(O, \rho) \cup \{w_1, \dots, w_s\}.$$

These recall the main-text proposition, whose proof is not repeated.

A.5. The local admissible set and radial envelope. Let u, v be unit vectors meeting at 120° and define

$$K(a, b, c) := \text{conv}\{0, au, bv, c(u+v)\}.$$

This is the translate of the body normalization at V_0 , with $u = V_5 - V_0$ and $v = V_1 - V_0$. The exact local admissible set is

$$(36) \quad \mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

This set records the three reach lower bounds that a closed unit V triangle can realize.

Proposition A.7 (Exact local admissible set). *Put*

$$\begin{aligned} s &:= a + b, & d &:= a^2 + ab + b^2, & m &:= \min\{a, b\}, & M &:= \max\{a, b\}, \\ q &:= s^4 - s^2 + ab, \end{aligned}$$

and

$$\begin{aligned} F_L &:= c^4 - c^2 + mc - m^2, \\ F_T &:= (s^2 - 1)c^2 + Mc - M^2, \\ F_S &:= (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Then

$$(37) \quad \mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S,$$

where all variables lie in $[0, 1]$ and

$$\begin{aligned}\mathcal{A}_L &:= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T &:= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S &:= \{d \leq 1, s > 1, F_S \leq 0, c \leq 1/2\}.\end{aligned}$$

The set is coordinatewise down-closed. Define its maximal radial demand by

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

Then

$$(38) \quad c_{\max}(a, b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \leq 1, q \leq 0, (a, b) \neq (0, 0), \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & s \leq 1, q > 0, \\ \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}, & s > 1, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

Proof. First assume $s > 0$ and $cs \leq ab$. When $a, b > 0$,

$$c(u + v) = \frac{c}{a}(au) + \frac{c}{b}(bv) + \left(1 - \frac{c}{a} - \frac{c}{b}\right)0 \in \text{conv}\{0, au, bv\};$$

the zero-coordinate cases follow by continuity. The distance from au to bv is $\sqrt{d}$. With $T = -bu - av$, the triangle $\text{conv}\{au, bv, T\}$ is equilateral of side $\sqrt{d}$ and

$$0 = \frac{b^2}{d}(au) + \frac{a^2}{d}(bv) + \frac{ab}{d}T.$$

Thus the least enclosure side is $\sqrt{d}$. When $a = b = 0$, it is c .

Now assume $cs > ab$. The hull has cyclic vertices $0, bv, c(u + v), au$. Lemma 5.2 reduces its least enclosure side to its four edge normals. For the edges through 0 one obtains

$$L_{0A} = b + \max\{a, c\}, \quad L_{B0} = a + \max\{b, c\}.$$

For the edge from au to $c(u + v)$ put

$$R_a := \sqrt{a^2 - ac + c^2}, \quad n_0 := \frac{1}{2R_a}(\sqrt{3}a, 2c - a).$$

The three support values in the rotated directions are

$$\frac{\sqrt{3}ac}{2R_a}, \quad \frac{\sqrt{3}a(a - c)_+}{2R_a}, \quad \frac{\sqrt{3}c \max\{b, (c - a)_+\}}{2R_a}.$$

Consequently

$$L_{AC} = \begin{cases} (a^2 + bc)/R_a, & a \geq c, \\ c(a + b)/R_a, & a < c \leq a + b, \\ c^2/R_a, & c > a + b, \end{cases}$$

and L_{CB} is obtained by interchanging a, b . Hence

$$\Lambda(K(a, b, c)) = \min\{L_{0A}, L_{B0}, L_{AC}, L_{CB}\}.$$

Assume by symmetry that $a = m \leq b = M$. Reading the active support points in these four formulas gives

range		active supports in cyclic order			unit frontier
s	q	first	second	third	equation
$s < 1$	$q < 0$	$\{0\}$	$\{c(u+v)\}$	$\{au, c(u+v)\}$	$F_L = 0$
$s < 1$	$q > 0$	$\{0\}$	$\{bv, c(u+v)\}$	$\{au\}$	$F_T = 0$
$s = 1$		$\{0, bv\}$	$\{bv, c(u+v)\}$	$\{au\}$	$c = M$
$s > 1$		$\{bv\}$	$\{bv, c(u+v)\}$	$\{au\}$	$F_S = 0$

The inactive-support inequalities reduce to $q \leq 0$, $q \geq 0$, and $s > 1$. The positive unit-frontier equations square to $F_L = 0$, $F_T = 0$, $F_S = 0$.

It remains to select the geometric components introduced by squaring. In Cell T the roots are

$$c_- := \frac{2M}{1 + \sqrt{4s^2 - 3}}, \quad c_+ := \frac{2M}{1 - \sqrt{4s^2 - 3}}.$$

The polynomial inequality also has a formal high component, while the geometric fiber is the component joined to $c = 0$. Since

$$c_- \leq 2M \leq c_+,$$

the exact selector is $c \leq 2M$.

In Cell S ,

$$F_S(a, b, 0) = M^2(M^2 - 1) < 0, \quad F_S(a, b, M) = M^2(s^2 - 1) > 0.$$

Moreover $4F_S(a, b, 1/2)$ increases with m , and at $m = 1 - M$ equals $M^2(2M - 1)^2$. Because $s > 1$, one has $F_S(a, b, 1/2) > 0$. The quadratic opens downward, so $c \leq 1/2$ selects exactly the smaller-root component containing 0.

In Cell L , the fiber from $c = 0$ reaches one root in $[\sqrt{3}/2, 1]$; uniqueness there follows from $4c^3 - 2c + m > 0$. At $q = 0$ the L and T formulas meet at $c = s$. This proves (37), including all equality and degenerate cases by continuity. Convexity of any containing triangle proves coordinatewise down-closedness. Solving the three selected frontier equations gives (38). $\square$

A.6. Self-midpoint corollary.

Lemma A.8 (Self-midpoint admissibility corollary). *If $a + b > 1$, then*

$$\Lambda\left(\text{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}\right) > 1.$$

This supplies Lemma 2.14.

Proof. Put $m = \min\{a, b\}$ and $M = \max\{a, b\}$. If the four-point hull were admissible with $a + b > 1$, Cell S of proposition A.7 would require $F_S(m, M, 1/2) \leq 0$. However,

$$4F_S(1 - M, M, 1/2) = M^2(2M - 1)^2 \geq 0,$$

$$\frac{\partial}{\partial m}(4F_S(m, M, 1/2)) = 2m + 4M^2 > 0.$$

Since $m > 1 - M$, this is impossible. If $a + b = 1$ and the four points have positive margin, increase the two boundary demands slightly to obtain the same contradiction. $\square$

Lemma A.9 (Strict-supercritical outgoing envelope). *If a V triangle is supercritical and has own-radial reach at least $c \in [0, 1/2)$, then both boundary reaches are strictly below M_c^{sup} from (16). No supercritical triangle has own-radial reach at least $1/2$.*

Proof. Let m and M be the smaller and larger actual boundary reaches. For $c > 0$, the selected supercritical cell gives $F_S(m, M, c) \leq 0$, whereas $m > 1 - M$ and

$$\partial_m F_S = 2c(M^2 + cm) > 0,$$

$$F_S(1 - M, M, c) = M(c - M)(1 - 2c + cM - M^2).$$

Since $M > 1/2 > c$, these identities imply $1 - 2c + cM - M^2 > 0$. Solving the quadratic gives $M < (c + \sqrt{c^2 - 8c + 4})/2$. For $c = 0$, each boundary reach is strictly less than one because its opposite distinguished vertex is excluded from the closed V triangle. The endpoint $c = 1/2$ is excluded by the self-midpoint obstruction. $\square$

A.7. Signed-center optimization. The signed-center coordinates describe a triangle that contains O in its interior and has a positive-length boundary trace. Their purpose is to relate its boundary endpoints to its radial exits, not to classify a new covering configuration. The same formulas will be applied later to an arbitrary enclosing candidate, with its parameters recomputed. The midpoint identity and the boundary-length cap are the geometric conclusions to retain from this calculation.

The CE1 and CE2 cases share one signed cell. The parameters below are used only in the appendices; the geometric body uses the exact C triangle type and midpoint interfaces.

Proposition A.10 (Signed CE1/CE2 center normal form). *Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$\begin{aligned} 0 < R < 1, \quad W &:= 1 - R, \quad E := \sqrt{1 - RW}, \\ \eta &:= 1 - E, \quad P := E(1 - E), \end{aligned}$$

and nonnegative slacks α, δ , with $k := \eta + \alpha + \delta$, such that

$$\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ (39) \quad F_1 &= Rb + Wa - k, \quad T_C = \{F_0, F_1, F_2 \geq 0\}. \\ F_2 &= W + \delta - b + Ra, \end{aligned}$$

Define

$$(40) \quad \Delta_R := P - \alpha - W\delta, \quad \Delta_L := P - R\alpha - \delta.$$

Then

$$(41) \quad T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta \right], \quad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R},$$

and the possible companion trace is

$$(42) \quad T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha \right] \quad \text{when } \Delta_L > 0, \quad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}.$$

The normalized trace requires $\Delta_R > 0$, and

$$(43) \quad T_C \text{ is CE1} \iff \Delta_L \leq 0, \quad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six radial exits from O are

$$(44) \quad \begin{aligned} d_0^C &= E - \alpha - \delta, & d_1^C &= \frac{\delta}{R}, & d_2^C &= \delta, \\ d_3^C &= \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}, & d_4^C &= \alpha, & d_5^C &= \frac{\alpha}{W}. \end{aligned}$$

Moreover,

$$(45) \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

For the closure of an original open C triangle, $\alpha, \delta > 0$.

Proof. Write $T_C \cap e_{0,1} = [s, t]$ with $0 \leq s < t \leq 1$. The two sides through the endpoints have a positive slope ratio. Reflect across the perpendicular bisector if necessary, call the ratio $R \leq 1$, and put $W = 1 - R$. The three side equations are

$$F_1 = Rb + Wa - Rs, \quad F_2 = -b + Ra + t, \quad F_0 = Wb - a + E + Rs - t,$$

where $E = \sqrt{1 - RW}$ is the unit-side relation. If $R = 1$, then $F_0(O) = s - t < 0$, so $0 < R < 1$.

Set $\alpha := F_0(O)$ and $\delta := F_2(O)$. Then

$$t = W + \delta, \quad Rs = \eta + \alpha + \delta = k,$$

which gives (39). The gradients are the three equilateral side-normal directions and $F_0 + F_1 + F_2 = E$.

On $e_{0,1}$, $a = 0$ and the active inequalities are $Rb \geq k$ and $b \leq W + \delta$. Thus its length is

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R}.$$

On $e_{5,0}$, $b = 0$ and the active inequalities are $Wa \geq k$ and $a \leq R + \alpha$, giving signed length

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W}.$$

This proves the two trace formulas.

The positive right trace implies

$$\delta < \frac{P}{W} = \frac{ER}{1+E} < \frac{R}{2}.$$

On $e_{1,2}$, parametrized by $b = 1 + q, a = q$, one has $F_2 = \delta - R - Wq < 0$. The center classification leaves $e_{5,0}$ as the only possible companion edge, and its signed length proves (43); $\Delta_L = 0$ is only point contact.

Substituting the six ray parametrizations into (39) gives (44). Finally,

$$W(\alpha + \delta) \leq \alpha + W\delta < P$$

implies

$$E - \alpha - \delta > E - \frac{P}{W} = \frac{E(E - R)}{W} > \frac{1}{2},$$

where the last inequality is equivalent to $(E - R)^2 > 0$. Also

$$\alpha < P < \min \left\{ \frac{R}{2}, \frac{W}{2} \right\}, \quad \delta < \frac{R}{2}.$$

Therefore $d_0^C > 1/2$ and $d_i^C < 1/2$ for $i = 1, \dots, 5$, proving the midpoint formula. Interior containment of O in the original open triangle makes all three side slacks positive, so $\alpha, \delta > 0$. $\square$

Corollary A.11 (Common center-boundary formula). *The full center-boundary contribution is*

$$(46) \quad L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In CE1,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4},$$

and in CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

Proof. The first formula sums (41) and (42). In CE1, $R\alpha + \delta \geq P$; multiplication by W , together with $\alpha \geq RW\alpha$, gives $\alpha + W\delta \geq WP$. Thus $\Delta_R \leq RP$ and the contribution is at most P . Since $E \geq \sqrt{3}/2$ and $E(1-E)$ decreases there, the stated cap follows.

In CE2, direct addition gives

$$L_{\partial H}(T_C) = \frac{P - E^2(\alpha + \delta)}{RW} = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

□

Lemma A.12 (Adjacent-edge diameter transfer). *For $0 \leq q \leq 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

Here M_0 without an argument remains the radial midpoint. If a unit equilateral triangle reaches parameters q, b on adjacent boundary edges, then $b \leq M_0(q)$. The function is strictly decreasing and, for $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \quad 1 - M_0(q) > \frac{q}{2}.$$

Proof. The squared distance between the boundary points is $q^2 + qb + b^2 \leq 1$. Solving the quadratic in b gives the formula. Differentiation gives

$$M'_0(q) = \frac{-1 - 3q/\sqrt{4 - 3q^2}}{2} < 0.$$

For $q > 0$, $\sqrt{4 - 3q^2} < 2$, which gives both strict comparisons. □

APPENDIX B. TRACE-LENGTH OPTIMIZATION

This appendix evaluates the trace quantities whose exact interfaces appear in Section 3. The center and V triangle coordinates are used only here; the body retains the geometric hypotheses and final bounds.

B.1. Vd1 and Vd2 corner optimization. Use temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$

so $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$.

Proposition B.1 (Vd1/Vd2 corner normal form). *Let T be the closure of an original Vd1 or Vd2 triangle at V_0 , and let A_0, B_0 be its exact reaches toward V_5, V_1 . There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$(47) \quad T = \left\{ (x, y) : \begin{array}{l} x - (t+1)y \leq A_0, \\ ty - (t+1)x \leq tB_0, \\ tx + y \leq d - A_0 - tB_0 \end{array} \right\}.$$

The raw traces, parametrized from the indicated outer vertex, are

$$(48) \quad \begin{array}{l} r_0 \\ r_1 \\ r_5 \end{array} \left| \begin{array}{l} 0 \leq s \leq \frac{d - A_0 - tB_0}{t+1} \\ \frac{t(1-B_0)}{t+1} \leq s \leq \frac{d - A_0 - tB_0 - 1}{t} \\ \frac{1-A_0}{t+1} \leq s \leq d - A_0 - tB_0 - t. \end{array} \right.$$

Intersecting with $[0, 1]$ gives the actual traces, and

$$(49) \quad A_0 + B_0 < \frac{1}{2}.$$

Proof. The triangle has one vertex W outside H and two vertices U, Z in H . Because V_0 is an interior convex combination, both coordinates of W are negative. The axis endpoints $(A_0, 0), (0, B_0)$ lie on the two sides through W . Put $p = U - W$ and $q = Z - W$. Their order selects

$$q = (p_x - p_y, p_x), \quad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \quad q = \frac{(t, t+1)}{d},$$

which yields (47); substitution of $(s, s), (s, 1), (1, s)$ yields (48).

If r_1 has positive trace, clearing its endpoint inequality gives

$$(t+1)A_0 + tB_0 < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(A_0 + B_0)$,

$$A_0 + B_0 < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

The last inequality follows from

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2 + t + 1) = \frac{t^2}{4} > 0.$$

Reflection treats support on r_5 . □

Lemma B.2 (Orientation of a forward Vd1 supplier). *In proposition B.1, if a Vd1 role contains M_1 openly, then $t \geq 1$.*

Proof. Suppose $t < 1$. Open containment of $M_1 = (1/2, 1)$ gives $A_0 + tB_0 < d - 1 - t/2$. The backward radial interval has endpoints $\ell_5 = (1 - A_0)/(t+1)$ and $u_5 = d - A_0 - tB_0 - t$, so

$$u_5 > 1 - t/2 > \frac{1}{t+1} > \ell_5, \quad 0 < \ell_5 < 1.$$

Here $1 - t/2 - 1/(t+1) = t(1-t)/(2(t+1)) > 0$. Thus r_5 has positive trace as well as r_1 , contradicting Vd1. $\square$

Lemma B.3 (Vd2 adjacent-midpoint cap). *If an original open V triangle has Vd2 closure T_i and*

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset,$$

then its exact boundary reaches satisfy

$$A_i + B_i < \frac{1}{3}.$$

Proof. In Proposition B.1, reflection sends

$$(t, A_0, B_0, M_1, M_5) \mapsto (1/t, B_0, A_0, M_5, M_1).$$

Thus assume $t \geq 1$ and use $A_0 + B_0 \leq A_0 + tB_0$.

If $M_5 = (1, 1/2)$ lies in T , the third half-plane gives

$$A_0 + B_0 \leq A_0 + tB_0 \leq d - t - \frac{1}{2} \leq \sqrt{3} - \frac{3}{2} < \frac{1}{3}.$$

If $M_1 = (1/2, 1)$ lies in T , the second and third half-planes give

$$B_0 \geq \frac{t-1}{2t}, \quad A_0 + B_0 \leq S(t) := d - t - \frac{1}{2t}.$$

Positive-length contact with r_5 gives

$$A_0 + B_0 < R(t) := \frac{2(t+1)d - 3t^2 - t - 2}{2t}.$$

Direct differentiation shows S increasing and R decreasing on $[1, \infty)$. For S' , the sign reduces to

$$(2t+1)^2 t^4 - (t^2 + t + 1)(2t^2 - 1)^2 = (t+1)(t^3 + 3t^2 - 1) > 0,$$

while $R' < 0$ follows from $d > t + 1/2$. Splitting at $t = 4/3$ gives

$$\begin{aligned} 1 \leq t \leq \frac{4}{3} : \quad A_0 + B_0 &\leq \frac{8\sqrt{37} - 41}{24} < \frac{1}{3}, \\ t \geq \frac{4}{3} : \quad A_0 + B_0 &< \frac{7\sqrt{37} - 39}{12} < \frac{1}{3}. \end{aligned}$$

The final comparisons are $8\sqrt{37} < 49$ and $7\sqrt{37} < 43$. $\square$

B.2. Conditional skeleton caps.

Lemma B.4 (Center-skeleton calculation). *Under the hypotheses of Lemma 3.2, one has $L_S(T_C) < 3/2$.*

Proof. Normalize the unique midpoint as M_0 and a positive boundary trace to $e_{0,1}$. Use the signed parameters and six radial exits of proposition A.10. The positive trace condition is

$$(50) \quad \alpha + W\delta < P.$$

Summing the boundary and radial contributions gives

$$L_S(T_C) \leq K_R + \mathcal{E}(\alpha, \delta), \quad K_R := W + E - \frac{\eta}{R},$$

where

$$\mathcal{E}(\alpha, \delta) := -\frac{\alpha}{R} + \frac{\alpha}{W} + \delta + \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\} + \frac{[P - (R\alpha + \delta)]_+}{W}.$$

Splitting according to the minimum and positive part gives

$$(51) \quad \mathcal{E}(\alpha, \delta) \leq \frac{P}{W}.$$

If $\alpha/R \leq \delta/W$, the expression without the positive part is $\alpha/W + \delta$; when that part is active, the excess over P/W is $\alpha - R\delta/W \leq 0$. The reflected order gives $\delta - W\alpha/R \leq 0$. If the positive part vanishes, (50) gives the same bound strictly.

Put $A := 1 + R - R^2$. Then

$$\frac{3}{2} - K_R - \frac{P}{W} = \frac{1 + A - 2EA}{2RW}.$$

Both sides of $1 + A > 2EA$ are positive, and

$$(1 + A)^2 - 4A^2E^2 = R^2W^2(5 + 4R - 4R^2) > 0.$$

Therefore $K_R + P/W < 3/2$, as required. $\square$

For the remaining cap, reflect the local directions if necessary so that $0 \leq A_i \leq B_i \leq 1$. The actual reach triple of a supercritical V triangle lies in the selected cell

$$(52) \quad \begin{aligned} A_i^2 + A_iB_i + B_i^2 &\leq 1, \\ C_i &\leq \frac{1}{2}, \\ (A_i^2 - 1)C_i^2 + (2A_iB_i^2 + B_i)C_i + (B_i^4 - B_i^2) &\leq 0, \end{aligned}$$

where C_i belongs to the connected component containing 0 in the corresponding one-dimensional fiber. This is Cell S of Proposition A.7. The selector is essential: the quadratic is negative at $C_i = 0$, positive at $C_i = 1/2$, and opens downward.

Lemma B.5 (Supercritical skeleton calculation). *Every V triangle with $A_i + B_i > 1$ has $L_S(T_i) < 3/2$.*

Proof. The corner normal form, Proposition B.1, makes Vd1 and Vd2 triangles nonsupercritical. Lemma 2.11 gives the same conclusion for T3-like triangles. Hence a supercritical triangle is Vd0 and has no positive adjacent-radial support. Reflect so that $A_i \leq B_i$. Then

$$L_S(T_i) = A_i + B_i + C_i.$$

Set

$$s := A_i + B_i, \quad \delta := B_i - A_i, \quad \gamma_0 := \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \leq \frac{2}{\sqrt{3}}, \quad 0 \leq \delta \leq \sqrt{4 - 3s^2}.$$

Let $P(\gamma)$ be the quadratic in (52), with $A_i = (s - \delta)/2$, $B_i = (s + \delta)/2$. Define $Q(s, \delta) := 16P(3/2 - s)$ by this exact substitution. Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)(\delta^2 + \delta(4s - 3) + (s - 1)(2 - s)) \geq 0.$$

Therefore

$$Q(s, \delta) \geq Q(s, 0) = (s - 1)(s^3 - 5s^2 - 24s + 36) > 0.$$

The cubic decreases on $[1, 2/\sqrt{3}]$ and at the right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(\gamma_0) > 0$. The selected smaller-root component in (52) forces $C_i < \gamma_0$, so $A_i + B_i + C_i < 3/2$. $\square$

APPENDIX C. AREA-LOSS OPTIMIZATION

This appendix proves the pointwise local loss bound of theorem 4.1. The cyclic accumulation is proved in the body.

C.1. Local square-loss optimization.

Lemma C.1 (Solved local square-loss optimization). *Every triangle satisfying the hypotheses of theorem 4.1 obeys equations (10) and (11).*

Proof. Use the shared corner chart of lemma A.2 and the orientation forms equations (24) and (25). Write $D = 1 - t + t^2$, $z = \sqrt{D}$, with $0 \leq t \leq 1$. In Type I, containment of $(0, 0), (0, a), (b, 0)$ gives

$$(53) \quad \alpha \geq tb, \quad \beta \geq (1 - t)a, \quad \alpha + \beta + (1 - t)b \leq z, \quad \alpha + \beta + ta \leq z.$$

The two inequalities defining W become

$$(1 - t)U + V \geq (1 - t)\alpha + \beta, \quad U + tV \geq \alpha + t\beta.$$

For fixed t , these inequalities describe the image of $T \cap W$ in the fixed simplex $\{U, V \geq 0, U + V \leq z\}$. Lowering α or β enlarges that image, not necessarily the physical intersection $T \cap W$. The Jacobian magnitude D is independent of these slacks, so the retained Euclidean area is nondecreasing. Thus the outside loss is minimized at $\alpha = tb$, $\beta = (1 - t)a$. At that placement the portion outside W has vertices, in the (U, V) -plane,

$$(0, 0), \quad (a + tb, 0), \quad (tb, (1 - t)a), \quad (0, b + (1 - t)a).$$

Since the Jacobian of $(x, y) \mapsto (U, V)$ has absolute value D ,

$$(54) \quad \mathcal{L}(T) \geq \mathcal{L}_I(a, b; t) = \frac{(1 - t)a^2 + tb^2 + 2t(1 - t)ab}{D}.$$

If $a \leq b$, then

$$D(\mathcal{L}_I - a^2) = t\{b^2 - a^2 + (1 - t)(a^2 + 2ab)\} \geq 0;$$

the reflected factorization proves $\mathcal{L}_I \geq b^2$ when $b \leq a$. This proves (10) in Type I.

Suppose now that $a + b > 1$ and $a \leq b$. Set

$$r = \frac{a}{b}, \quad t_0 = \frac{1 - r^2}{1 + 2r}.$$

Since $b > 1/(1 + r)$, feasibility in (54) and (53) excludes $0 \leq t < t_0$: for $0 < t < t_0$,

$$\left(\frac{1 + r(1 - t)}{1 + r}\right)^2 - D = \frac{t(1 + 2r)(t_0 - t)}{(1 + r)^2} > 0,$$

so $b + (1 - t)a > z$, while $t = 0$ would give $a + b \leq 1$. Hence $t \geq t_0$, and

$$D(\mathcal{L}_I - b^2) = (1 - t)b^2\{r^2 - 1 + t(2r + 1)\} \geq 0.$$

If $b \leq a$, put $r = b/a$ and $t_1 = (r^2 + 2r)/(1 + 2r)$. The identical reflected calculation excludes $t > t_1$ and gives

$$D(\mathcal{L}_I - a^2) = ta^2\{r^2 + 2r - t(1 + 2r)\} \geq 0.$$

Thus (11) holds in Type I.

In Type II, substituting the axes in equation (25) gives the exact positive y - and x -axis reaches

$$(55) \quad A = \beta, \quad B = z - \alpha - \beta, \quad A + B \leq z \leq 1.$$

In particular Type II is impossible when $a + b > 1$. Its wedge intersection is the quadrilateral with vertices

$$(0, 0), \quad (B, 0), \quad \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D} \right), \quad (0, A).$$

Hence

$$(56) \quad \mathcal{L}(T) = \mathcal{L}_{\text{II}}(A, B; t) = 1 - \frac{(1-t)A^2 + tB^2 + 2AB}{D}.$$

Let $S = A + B$. If $A \leq B$, write $A = rS$, $B = (1-r)S$ with $0 \leq r \leq 1/2$. Since $S^2 \leq D$, and with

$$C = (1-t)r^2 + t(1-r)^2 + 2r(1-r),$$

we have $\mathcal{L}_{\text{II}} \geq 1 - C$. The exact factorization

$$1 - C - r^2D = (1-t)(1 - 2r + tr^2) \geq 0$$

gives $\mathcal{L}_{\text{II}} \geq r^2D \geq A^2$. Reflection gives $\mathcal{L}_{\text{II}} \geq B^2$ when $B \leq A$. Since $A \geq a$ and $B \geq b$, this proves (10). The two orientation types are exhaustive, so the theorem follows. $\square$

APPENDIX D. NONZERO-GAP FINITE-ENCLOSURE OPTIMIZATION

This appendix proves the selected-gap and supported-rescuer enclosure statements and the two-vertex replacement. The witnesses are fixed in Section 5; the terminal enclosure proofs use hull-edge calipers rather than signed center parameters.

D.1. Direct neighboring-ray domination. The capacities keep their variational definitions. The following proof establishes the bound needed by all active witnesses without evaluating the cubic plateau and radical branches of the full function.

Lemma D.1 (Sharp diagonal neighboring capacity). *For $0 \leq m \leq 1/2$, one has $C_+(m, m) = C_-(m, m) = 1 - m$.*

Proof. Use oblique coordinates with metric $x^2 + y^2 - xy$. The four points are $P = (0, 0)$, $A = (m, 0)$, $B = (0, m)$, and $Z = (c, 1)$. The case $m = 0$ is immediate. If $0 < m \leq 1/2$ and $c > 1 - m$, their hull has cyclic order P, A, Z, B . lemma 5.2 reduces the enclosure to its four outward edge normals. At PA and BP , the side lengths are $c + m$ and $1 + m$, both greater than one.

Embed (x, y) as $(x - y/2, hy)$. At the other two edges take unnormalized normals $n_A = (h, 1/2 - c + m)$ and $n_B = (h(m - 1), m/2 + c - 1/2)$. Put

$$\begin{aligned} D_A &= (c - m)^2 - (c - m) + 1, & N_A &= m + c^2 - cm - c + 1, \\ D_B &= c^2 + cm - c + m^2 - 2m + 1, & N_B &= cm + c^2 - c - m + 1. \end{aligned}$$

Selecting the contained points (A, Z, P) in the three normal directions at n_A , and (B, P, Z) at n_B , gives support lower bounds $L_{AZ} \geq N_A/\sqrt{D_A}$ and $L_{ZB} \geq N_B/\sqrt{D_B}$. No active-support selector is assumed. With $q = c + m - 1 > 0$,

$$\begin{aligned} N_A^2 - D_A &= q^2((q + 1 - 3m)^2 + 4m^2 - 2m + 1) \\ &\quad + q(1 - 2m)(6m^2 - 2m + 1) + m^2(1 - 2m)^2 > 0, \\ N_B^2 - D_B &= q^4 + (2 - 2m)q^3 + (m^2 - 4m + 2)q^2 \\ &\quad + (1 - m)(1 - 2m)q > 0. \end{aligned}$$

Here $4m^2 - 2m + 1 \geq 3/4$, $6m^2 - 2m + 1 > 0$, and $m^2 - 4m + 2 \geq 1/4$. The numerators are positive: $N_A \geq (1+m)(3-m)/4$ and $N_B = 1 - m + (1-m)q + q^2$. Thus every caliper requires side greater than one. Equality is realized by the unit triangle $\text{conv}\{(-m, 0), (1-m, 0), (1-m, 1)\}$. Reflection gives C_- . $\square$

D.2. Elementary local capacity bounds. Write $f(a, b) = 1 - c_{\max}(a, b)$ for $a, b \geq 0$, $a + b \leq 1$, and put $m = \min(a, b)$. Let $h = \sqrt{3}/2$ and let $c(m) \in [h, 1]$ be the root of

$$F(C, m) = C^4 - C^2 + mC - m^2 = 0, \quad \ell(m) = 1 - c(m).$$

It is unique because $F_C > 0$ on $[h, 1]$, $F(h, m) = -(m - h/2)^2 \leq 0$, and $F(1, m) = m(1 - m) \geq 0$. Define

$$q(m) = \frac{m(1-m)}{2}, \quad \beta(m) = \frac{2m}{4+3m}, \quad \kappa(m) = 1 + \frac{3}{2}m.$$

Lemma D.2 (Own-ray deficit bounds). *For $a, b \geq 0$, $a + b \leq 1$, and $m = \min(a, b)$,*

$$q(m) \leq \ell(m) \leq f(a, b) \leq m.$$

For $m \leq 1/4$, also $\beta(m) \leq \ell(m)$ and $\beta(m) \geq q(m)$.

Proof. The upper bound and capacity antitonicity follow from lemma 5.5. By the selected local formula proposition A.7, for $m \leq h/2$ the diagonal pair lies in the quartic cell (its cell polynomial is $m^2(16m^2 - 3) \leq 0$), so $c_{\max}(a, b) \leq c_{\max}(m, m) = c(m)$. For $m \geq h/2$, antitonicity gives $c_{\max}(a, b) \leq c_{\max}(h/2, h/2) = h \leq c(m)$. Thus $f(a, b) \geq \ell(m)$.

For $c = c(m)$, the root equation and a factorization give

$$\ell c^2(1+c) = m(c-m),$$

$$2(c-m) - (1-m)c^2(1+c) = (1-c)((1-m)c(c+2) - 2m) \geq 0.$$

The bracket is at least $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{11}{4} - 1 = 1/32$, since $c \geq h > 3/4$ and $m \leq 1/2$. Division by $c^2(1+c) > 0$ proves $\ell \geq q(m)$.

For the stronger bound on $m \leq 1/4$, put

$$B(m) = 81m^4 + 405m^3 + 656m^2 + 272m - 128.$$

Direct substitution gives

$$F(1 - \beta(m), m) = -\frac{m^2 B(m)}{(4+3m)^4}.$$

The polynomial B is increasing and $B(1/4) = -3163/256 < 0$. Also $1 - \beta(m) \geq 17/19 > h$. Monotonicity of F in C implies $\beta(m) \leq \ell(m)$. Finally $\beta(m) - q(m) = m^2(3m+1)/(2(3m+4)) \geq 0$. At $m = 0$ all bounds are immediate. $\square$

Lemma D.3 (Quarter-range slack envelope). *For $a, b \geq 0$, $a + b \leq 1$, $m = \min(a, b) \leq 1/4$, and $\delta = 1 - a - b$,*

$$f(a, b) \geq \max\{\beta(m), m - \kappa(m)\delta\}.$$

The two entries meet at $\delta = \beta(m)$. This estimate is asserted only on $m \leq 1/4$; the broader historical envelope is not needed here.

Proof. Since $m - \kappa(m)\beta(m) = \beta(m)$, the baseline in lemma D.2 proves the claim for $\delta \geq \beta(m)$. The case $m = 0$ is immediate. Suppose $m > 0$ and $0 < \delta \leq \beta(m)$. Put $M = 1 - m - \delta$. Then $M \geq m$ (indeed $1 - 2m - \beta(m) \geq 15/38 > 0$) and $1 - \delta \geq 1 - \beta(m) \geq c(m)$. The selected triangular branch of proposition A.7 therefore applies: the capacity is the smaller root of

$$H(C) = ((1 - \delta)^2 - 1)C^2 + MC - M^2.$$

At $\bar{C} = 1 - m + \kappa(m)\delta$, expansion gives $H(\bar{C}) = \delta N_m(\delta)/4$, where

$$\begin{aligned} N_m(\delta) &= (3m + 2)^2\delta^3 - 10m(3m + 2)\delta^2 \\ &\quad + (28m^2 - 22m - 20)\delta + 14m(1 - m). \end{aligned}$$

Because $\delta \leq \beta(m) \leq m/2 \leq 1/8$,

$$N'_m(\delta) \leq 3(11/4)^2(1/8)^2 + 7/4 - 20 = -18325/1024 < 0.$$

At the right endpoint,

$$N_m(\beta(m)) = -\frac{2mB(m)}{(4 + 3m)^3} > 0.$$

Thus $H(\bar{C}) > 0$. The quadratic is concave and the capacity is its selected smaller root, so $c_{\max}(M, m) < \bar{C}$. This proves the affine lower bound. At $\delta = 0$, the exact capacity is $1 - m$. $\square$

Lemma D.4 (Coupled nonuniform capacity inequality). *If $0 < x \leq y < 1$ and $x/2 \leq z \leq y$, put*

$$r = f(1 - y, z), \quad t = f(1 - z, x/2).$$

Then

$$2(r + t) > (1 - y + r)(2x - y + r).$$

Proof. Put $u = x/2$ and

$$P(y, r, t) = 2(r + t) - (1 - y + r)(4u - y + r).$$

The domain is $0 < u, 2u \leq y < 1, u \leq z \leq y$ and $0 < r, t \leq 1/2$. By lemma D.2,

$$P_r = 1 - 4u + 2y - 2r \geq 1 - 2r \geq 0, \quad P_t = 2.$$

Hence valid lower bounds on either radius may be substituted. These are scalar comparisons; the exact witness radii are not changed.

Easy regions. First let $y \geq 1/2$ and put $a = 1 - y$. Then $0 < a \leq 1/2$, $u \leq (1 - a)/2$, and $u \leq z \leq 1 - a$. If $z \geq a$ and $z \leq 1 - u$, substitute $q(a), q(u)$. The expression is concave in u ; at $u = 0$ it equals $2q(a) + (a + q(a))(1 - a - q(a)) > 0$, and at $u = (1 - a)/2$ it equals $(1 - a)^3(1 + a)/4 > 0$. If $z \geq a$ and $z \geq 1 - u$, both radii are at least $q(a)$ because $a \leq 1 - z \leq u \leq 1/2$ and q is increasing. The substituted expression decreases with u and at $u = (1 - a)/2$ equals $a(1 - a)(a^2 - a + 2)/4 > 0$. If $z \leq a$, both radii are at least $q(u)$. The expression is concave in $a \in [u, \min(1/2, 1 - 2u)]$, with endpoint values

a	P after substitution	u range
u	$u(14 - 43u + 14u^2 - u^3)/4$	$(0, 1/3]$
$1/2$	$(1 - 17u^2 + 10u^3 - u^4)/4$	$(0, 1/4]$
$1 - 2u$	$u(-u^3 + 2u^2 + 9u - 2)/4$	$[1/4, 1/3]$

The three brackets respectively decrease to $32/27$, decrease to $23/256$, and increase from $23/64$, so all are positive.

Now let $y \leq 1/2$ and $z \geq 2u$. Substitute $q(z), q(u)$. The result increases with y , so set $y = z$; it is then concave in $u \in [0, z/2]$. The value at zero is positive, and the value at $u = z/2$ is $z^3(2-z)/4 > 0$. Finally, for $y \leq 1/2$ and $1/4 \leq z \leq 2u$, the same substitution allows $y = 2u$, giving

$$P \geq q(z)(1 - q(z)) - u + 3u^2 \geq \frac{3}{32} \frac{29}{32} - \frac{1}{12} = \frac{5}{3072} > 0.$$

Here $q(z)(1 - q(z))$ increases on $[1/4, 1/2]$ and $-u + 3u^2 = 3(u - 1/6)^2 - 1/12$.

The remaining region and its switches. It remains to treat

$$0 < u \leq 1/4, \quad u \leq z \leq \min(2u, 1/4), \quad 2u \leq y \leq 1/2.$$

The smaller capacity coordinates are z and u . By lemma D.3 compare with

$$R_0 = \max\{\beta(z), z - \kappa(z)(y - z)\}, \quad T_0 = \max\{\beta(u), u - \kappa(u)(z - u)\}.$$

Put $g(v) = v + \beta(v)$, an increasing rational function. The switches are $y = g(z)$ and $z = g(u)$. For fixed u, z , the affine branch of R_0 gives

$$\frac{d^2}{dy^2} P(y, R_0, T_0) = -2(1 + \kappa(z))^2 < 0.$$

The baseline branch gives $dP/dy = 1 + 4u + 2\beta(z) - 2y > 0$. Since $g(z) \leq g(1/4) = 27/76 < 1/2$, only $y = 2u$ or $y = g(z)$ needs to be checked, whenever the latter belongs to the domain.

On $y = 2u$,

$$P = R_0(1 - R_0) + 2T_0 - 2u(1 - 2u).$$

Split the z interval at $g(z) = 2u$ and $z = g(u)$. The function $\beta(z)(1 - \beta(z))$ is concave, since β is concave and $v(1 - v)$ is increasing and concave on $[0, 1/2]$. On the other branch,

$$R = 2z + \frac{3}{2}z^2 - 2u - 3uz, \quad R' = 2 + 3z - 3u \geq 2, \quad R'' = 3.$$

Where it is active, $0 < R \leq z \leq 1/4$, so

$$(R(1 - R))'' = 3(1 - 2R) - 2(R')^2 \leq -5 < 0.$$

As T_0 is affine or constant on each piece, P is piecewise concave. At $z = u$ it equals $\beta(u)(1 - \beta(u)) + 4u^2 > 0$; at $z = 2u$ it is $2T_0 > 0$. The endpoint $z = 1/4$ is covered by the easy-region bound, since the comparison radii dominate the q bounds. Only the switches remain.

On $y = g(z)$,

$$P = 2\beta(z) + 2T_0 - (1 - z)(4u - z), \quad z/2 \leq u \leq g(z)/2.$$

Its u derivative on the baseline branch is $16/(4 + 3u)^2 - 4(1 - z) \leq -2 < 0$; on the other branch it is $6u + z > 0$. Thus the minimum lies at an endpoint or $z = g(u)$. The endpoints are $z = 2u$, already checked, and transition A below. The ordering of switches is not assumed; coincident switches are included.

Three transition checks. In A, $y = 2u = g(z)$. Using the affine entry as a lower bound for T_0 gives

$$P \geq \frac{z^2(9z^2 + 36z + 44)}{4(3z + 4)^2} > 0.$$

In B, $y = 2u$ and $z = g(u)$, so $T_0 = \beta(u)$. If R_B denotes the affine entry of R_0 , then

$$R_B - u(1 - 2u) = \frac{u^2(9u^2 + 6u + 4)}{2(3u + 4)^2} > 0.$$

Since $v(1 - v)$ increases on $[0, 1/2]$,

$$P \geq \frac{u^2(1 + 19u - 4u^2 - 12u^3)}{3u + 4} > 0,$$

where $4u^2 + 12u^3 \leq 7/16$. In C, $y = g(z)$ and $z = g(u)$, direct substitution gives

$$P = \frac{3u^2(81u^4 + 441u^3 + 768u^2 + 460u + 48)}{(3u + 2)(3u + 4)^2(3u + 8)} > 0.$$

These regions and transitions are exhaustive. Monotonicity in the radii therefore proves the inequality for the exact capacities. $\square$

D.3. A five-point threshold criterion.

Lemma D.5 (Five-point threshold criterion). *Suppose $0 < x \leq y < 1$, $0 < r \leq \min(y, 1 - y)$, and $0 < t \leq x/2$. Put $G_w = (1 - w)V_0 + wV_1$, $k = y - r$, and $F = \{M_0, G_x, G_y, rV_2, tV_4\}$. Then*

$$(57) \quad \Lambda(F) > 1 \iff r + t + \max\{1/2, 1 - x(1 - k)\} > \sqrt{1 - k + k^2}.$$

The gap points may coincide. This is a criterion at side one, not a claim that the displayed caliper always minimizes the enclosing side.

Proof. Put $M = M_0$, $P = rV_2$, and $Q = tV_4$. For $x < y$, these points have hull order M, G_x, G_y, P, Q . The stated bounds make the consecutive cross products positive. At G_yP , the outward normal $n = (-hk, 1 - k/2)$ has squared norm $1 - k + k^2$. The three supports divided by h are r , t , and $\max\{1/2, 1 - x(1 - k)\}$. Hence its caliper side is

$$L_3 = \frac{r + t + \max\{1/2, 1 - x(1 - k)\}}{\sqrt{1 - k + k^2}}.$$

The other four calipers exceed one. At the edges MG_x , G_xG_y , and QM , selected point projections give

$$L_1 > \frac{1 - x + 2x^2}{\sqrt{1 - 2x + 4x^2}}, \quad L_2 = 1 + r + t, \quad L_5 \geq \frac{1 + t + 2t^2}{\sqrt{1 + 2t + 4t^2}}.$$

For L_1 use $y \geq x$ and the strictly positive contribution $2tx$; for L_5 use $y \geq x \geq 2t$. The squared differences are

$$x^2(2x - 1)^2 \geq 0, \quad t^2(1 + 2t)^2 > 0,$$

respectively. At PQ , put $v = t/r > 0$. Selected projections give

$$L_4 \geq \frac{t + v + x + \max\{1/2, 1 - (1 + v)x\}}{\sqrt{1 + v + v^2}}.$$

The piecewise-linear expression in x is minimized at $1/[2(1 + v)]$. Thus its numerator exceeds $g(v) = v + 1/2 + 1/[2(1 + v)]$, and

$$g(v)^2 - (1 + v + v^2) = \frac{v^2}{4(1 + v)^2} > 0.$$

By theorem 5.3, only L_3 can cross the unit threshold. When $x = y$, discard the zero edge G_xG_y ; the other four calculations remain valid, without a limiting strictness argument. $\square$

Capacity instance. For $x/2 \leq z \leq y$, take $r = f(1 - y, z)$ and $t = f(1 - z, x/2)$. The bounds of lemma D.2 meet the geometric hypotheses, and lemma D.4 gives $r + t > (1 - k)(x - k/2)$. Consequently the left side of (57) exceeds $1 - k/2 + k^2/2 \geq \sqrt{1 - k + k^2}$; the squared difference is $k^2(1 - k)^2/4$. Thus this capacity instance has $\Lambda(F) > 1$.

D.4. The supported-rescuer adapter.

Lemma D.6 (Supported endpoint to four-point enclosure). *Suppose T_0 has actual boundary endpoint $a = A_0$ toward V_5 and supported interval $[c, u]$ on r_1 , measured from V_1 toward O . Put $\varepsilon = 1 - u > 0$ and $M = M_c^{\text{sup}}$. Assume*

$$\varepsilon V_1 \in U_C, \quad A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M.$$

If T_1 is uniquely supercritical with $C_1 \geq c$, and the four center-free edges from $e_{1,2}$ through $e_{4,5}$ join nonsupercritical T_2, T_3, T_4, T_5 , then the skeleton is not covered. The conclusion applies to both possible nonzero gap ranks.

Proof. The strict-supercritical envelope gives $B_1 < M$, and the four handoffs give $B_5 \leq B_4 \leq B_3 \leq B_2 \leq B_1$. Thus $B_5 < 1 - A_0$, so the left-edge actual gap has endpoints $Y(A_0), Y(1 - B_5)$, with $Y(t) = (1 - t)V_0 + tV_5$. Under coverage these endpoints and $O, \varepsilon V_1$ belong to U_C . Since $M \geq 1/2$, the hypotheses imply

$$A_0 \leq \varepsilon, \quad B_5 < M \leq \frac{\varepsilon}{A_0 + \varepsilon}.$$

Apply theorem 5.16 with geometric parameters $a = A_0, \beta = B_5$. The fixed set has enclosure number at least one, contrary to compact containment in U_C . $\square$

D.5. T3-like supported-tail adapter.

Lemma D.7 (Solved T3-like supported-tail adapter). *Assume a nonzero-gap skeleton cover has center midpoint M_0 , a T3-like role T_0 supplying M_1 , uniquely supercritical T_1 , and nonsupercritical $Vd0$ roles at the other four vertices. The actual supported interval satisfies the hypotheses of lemma D.6.*

Proof. Use the original Type-II chart (25), with positive first slack α_T and actual reach $a = A_0$. Thus $0 < t < 1, z = \sqrt{1 - t + t^2}$, and substitution on r_1 gives

$$(58) \quad c = \frac{1 + \alpha_T + a - z}{1 - t}, \quad u = a + t, \quad \varepsilon = 1 - u.$$

Open midpoint containment gives $0 < c < 1/2 < u < 1$. The supercritical own role ends before the midpoint and the other roles have no positive trace on r_1 . Thus u is the total radial frontier: $\varepsilon V_1 \in U_C$ by lemma 5.8. Coverage of the near endpoint gives $C_1 \geq c$, since the center exits before M_1 .

Put $x = a/(1 - t)$ and $\theta = t/(1 + z)$. Then

$$(59) \quad a + \varepsilon = 1 - t < 1, \quad c = x + \theta + \frac{\alpha_T}{1 - t} > x + \theta, \quad \frac{a}{a + \varepsilon} = x.$$

In particular $0 < x < c < 1/2$. Since $z < 1$ and $x + (1 - x)t = a + t > 1/2$,

$$\theta > \frac{t}{2} > \frac{1 - 2x}{4(1 - x)}.$$

The quadratic $Q_c(x) = x^2 + (c - 2)x + c$ increases with c , so

$$Q_c(x) > 2x^2 + (\theta - 1)x + \theta > \frac{(1 - 2x)(4x^2 - 3x + 1)}{4(1 - x)} > 0.$$

Here $4x^2 - 3x + 1 = 4(x - 3/8)^2 + 7/16$. The scalar rescuer test lemma 5.17 gives $x \leq 1 - M_c^{\text{sup}}$. Together with the actual size relation and the forced endpoint, these are exactly the common D hypotheses. No translated near endpoint is used. $\square$

Lemma D.8 (Solved Vd1 supported-tail adapter). *In the center-based Vd1 supplier placement, with $M_1 \in U_0$ and T_1 uniquely supercritical, the actual corner data imply the size and ratio inequalities required by lemma D.6.*

Proof. Set $a = A_0$, $b = B_0$. By lemma B.2, the original corner parameter satisfies $t \geq 1$. Put $d = \sqrt{t^2 + t + 1}$ and write the actual supported interval as $[c, u]$. Then $0 < c < 1/2 < u < 1$ and

$$c = \frac{t(1 - b)}{t + 1}, \quad u = \frac{d - a - tb - 1}{t}.$$

The midpoint condition and $tb = t - c(t + 1)$ imply

$$a \leq F(t, c) := d - 1 - \frac{3t}{2} + c(t + 1).$$

Since $d' < 1$ and $c < 1/2$, F decreases with t , so

$$a \leq F(t, c) \leq F(1, c) = \sqrt{3} - \frac{5}{2} + 2c =: L(c).$$

Put $\varepsilon = 1 - u$. Direct substitution gives

$$\varepsilon = \varepsilon_0 + \frac{a}{t}, \quad \varepsilon_0 = \frac{2t + 1 - c(t + 1) - d}{t} > 0.$$

For the last sign, $d < t + 1$ gives $t\varepsilon_0 > (1 - c)t - c \geq 1 - 2c > 0$. The function $v/(v + \varepsilon_0 + v/t)$ increases for $v \geq 0$, and $\varepsilon_0 + F(t, c)/t = 1/2$. Hence, with $x = a/(a + \varepsilon)$,

$$x \leq \frac{F(t, c)}{F(t, c) + 1/2} \leq 2F(t, c) \leq 2L(c) < 4c - \frac{3}{2},$$

using $\sqrt{3} < 7/4$. Thus $0 < x < 1/2$ and $c > (3 + 2x)/8$. Therefore

$$x^2 + (c - 2)x + c > x^2 + \left(\frac{3 + 2x}{8} - 2\right)x + \frac{3 + 2x}{8} = \frac{(1 - 2x)(3 - 5x)}{8} > 0.$$

The scalar rescuer test lemma 5.17 proves $x \leq 1 - M_c^{\text{sup}}$.

Positive forward support also gives

$$(t + 1)a + tb < (t + 1)(d - 1) - t^2 < d - 1,$$

because $d < t + 1$. Thus $u > a$ and $a + \varepsilon = 1 + a - u < 1$. As in the T3-like placement, u is the total frontier and the near endpoint is supplied by T_1 , so $\varepsilon V_1 \in U_C$ and $C_1 \geq c$. These are all the common D hypotheses. $\square$

D.6. Vd1 two-chart replacement. The replacement below uses two different vertex charts. It preserves the five skeleton pieces that can change, then invokes the all-nonsupercritical obstruction without assuming preservation of the input gap rank.

Lemma D.9 (Half-square admissibility). *If $(x, y, z) \in \mathcal{A}$ satisfies*

$$x \geq \frac{1}{2}, \quad 0 \leq z \leq \frac{1}{2},$$

then

$$y \leq 1 - z.$$

Proof. Suppose $y > 1 - z$. Then $x + y > 1$, so the selected supercritical cell of proposition A.7 applies. In the ordered half $x \leq y$ its necessary polynomial is

$$F(x, y, z) = (x^2 - 1)z^2 + (2xy^2 + y)z + y^4 - y^2 \leq 0.$$

It is nondecreasing in x , its derivative in y is positive for $y \geq 1 - z$, and

$$F\left(\frac{1}{2}, 1 - z, z\right) = \frac{z^2(2z - 5)(2z - 1)}{4} \geq 0.$$

This is a contradiction. In the reflected half $y < x$, apply the same argument to $F(y, x, z)$. At the joining corner,

$$F(1 - z, 1 - z, z) = z(1 - 2z) \geq 0,$$

so no selected feasible point is lost at equality. $\square$

Lemma D.10 (Two-vertex replacement from scalar margins). *Consider two adjacent V roles. Only the five pieces*

$$e_{5,0}, \quad e_{0,1}, \quad e_{1,2}, \quad r_0, \quad r_1$$

may be affected by their replacement, and the shared edge is center-free. Let $a, c, B, r \in [0, 1]$ be the boundary reach on $e_{5,0}$, the own-radial reach on r_0 , the boundary reach on $e_{1,2}$, and the radial demand on r_1 needed to overlap the unchanged center interval. If

$$a < \frac{1}{2}, \quad a + c < 1, \quad a + B < 1, \quad r < \max\{1 - a, 1 - B\},$$

there are two open unit equilateral replacements preserving these five skeleton pieces, with actual boundary sums below one and no positive adjacent support. No center-class formula is assumed.

Proof. Choose first $p_2 \in (a, 1 - B)$ with $\max\{p_2, 1 - p_2\} > r$. This is possible because the supremum on that interval is $\max\{1 - a, 1 - B\}$. Next choose $a < p_1 < \min\{p_2, 1/2, 1 - c\}$ and

$$0 < \varepsilon < \min\{p_1 - a, p_2 - p_1, 1 - B - p_2, 1 - p_1 - c, \max(p_2, 1 - p_2) - r\}.$$

All five margins are positive. Use the separate vertex charts

$$X_0(x, y) = V_0 + x(V_5 - V_0) + y(V_1 - V_0), \quad X_1(x, y) = V_1 + x(V_0 - V_1) + y(V_2 - V_1).$$

Both have metric $x^2 + y^2 - xy$. Define

$$\Delta_p^- = \text{conv}\{(0, 1 - p), (1, 1 - p), (0, -p)\}, \quad D_{p,\varepsilon}^- = \text{int}(\Delta_p^-) + (-\varepsilon, 0) \quad (p \leq 1/2),$$

$$\Delta_p^+ = \text{conv}\{(p, 0), (p, 1), (p - 1, 0)\}, \quad D_{p,\varepsilon}^+ = \text{int}(\Delta_p^+) + (0, -\varepsilon) \quad (p > 1/2).$$

Their edges have unit length. The minus half-planes are $x > -\varepsilon$, $y < 1 - p$, $y > x + \varepsilon - p$; the plus half-planes are $y > -\varepsilon$, $x < p$, $x > y + \varepsilon + p - 1$. Substitution on the axes and diagonal gives their actual reaches:

	A	B	C
$D_{p,\varepsilon}^-$	$p - \varepsilon$	$1 - p$	$1 - p$
$D_{p,\varepsilon}^+$	p	$1 - p - \varepsilon$	p

Both contain the chart origin. On the adjacent support lines $(z, 1)$ and $(1, z)$ the half-planes are incompatible for $0 \leq z \leq 1$, so neither has positive adjacent support. Set $U'_0 = X_0(D_{p_1,\varepsilon}^-)$ and use $U'_1 = X_1(D_{p_2,\varepsilon}^-)$ for $p_2 \leq 1/2$, or $U'_1 = X_1(D_{p_2,\varepsilon}^+)$ otherwise. The two charts and this split cannot be identified. Both actual boundary sums are $1 - \varepsilon$.

The first replacement reaches $p_1 - \varepsilon > a$ on $e_{5,0}$ and $1 - p_1 > c$ on r_0 . On the shared edge the reaches sum to at least

$$(1 - p_1) + (p_2 - \varepsilon) > 1,$$

so the open traces overlap. The second reaches more than B on $e_{1,2}$ and $\max(p_2, 1 - p_2) > r$ on r_1 . Thus all five pieces remain covered; all other pieces are unchanged. $\square$

Proposition D.11 (Vd1 input to the two-vertex replacement). *Assume a skeleton cover has a unique Vd1 role T_0 supplying M_1 , a unique supercritical role T_1 , and all other V roles are nonsupercritical Vd0. Suppose $e_{0,1}$ is center-free and no other V role has positive trace on r_0 or r_1 . The scalar replacement applies in CE1 and CE2 and yields a skeleton cover with $N'_+ = 0$.*

Proof. Let A_0, B_0, C_0 be the actual incoming, outgoing, and own-radial reaches of the Vd1 role. Its corner normal form and lemma B.2 supply $\vartheta \geq 1$ and

$$\omega := \sqrt{\vartheta^2 + \vartheta + 1}$$

such that

$$C_0 = \frac{\omega - A_0 - \vartheta B_0}{\vartheta + 1}, \quad \lambda = \frac{\vartheta(1 - B_0)}{\vartheta + 1},$$

$$u_{\text{adj}} = \frac{\omega - A_0 - \vartheta B_0 - 1}{\vartheta}.$$

The strict midpoint and one-sided-support conditions give

$$A_0 + C_0 < 1, \quad A_0 < \lambda \leq \frac{1}{2},$$

$$B_0 < \frac{1}{2}, \quad u_{\text{adj}} < 1 - A_0.$$

For the first inequality, evaluation on the closed midpoint face reduces strictness to

$$(2\vartheta^2 + 4\vartheta + 1)^2 - 4(\vartheta + 1)^2(\vartheta^2 + \vartheta + 1) = 4\vartheta^3 + 4\vartheta^2 - 4\vartheta - 3 > 0.$$

The remaining inequalities follow directly from the displayed endpoint formulas and the strict midpoint inequalities.

The center-free shared edge and the supported interval give

$$A_1 \geq 1 - B_0 > \frac{1}{2}, \quad C_1 \geq \lambda$$

for the actual reaches of T_1 . The selected supercritical component has $C_1 \leq 1/2$, so lemma D.9 yields

$$B_1 \leq 1 - C_1 \leq 1 - \lambda.$$

Since $A_0 < \lambda$,

$$A_0 + B_1 < 1.$$

Define the C-triangle reach on r_1 , measured from O , by

$$d_1^C := \max\{x \in [0, 1] : xV_1 \in T_C\}, \quad c_1^{\text{req}} := 1 - d_1^C.$$

Only U_C, U_1 , and the adjacent trace of U_0 can contribute a positive interval to r_1 . Open skeleton coverage gives

$$c_1^{\text{req}} < \max\{C_1, u_{\text{adj}}\} \leq \max\{1 - B_1, 1 - A_0\}.$$

Apply lemma D.10 with

$$(a, c, B, r) = (A_0, C_0, B_1, c_1^{\text{req}}).$$

All four scalar conditions were proved above. The four untouched roles remain nonsupercritical Vd0, while the two new actual boundary sums are $1 - \varepsilon < 1$. The resulting skeleton cover contradicts N0. The original C triangle is fixed; its class and the input gap rank were not used in the construction. This operation does not claim to preserve coverage of the interior of H . $\square$

APPENDIX E. ZERO-GAP NINE-POINT OPTIMIZATION

This appendix proves the frontier construction, witness exclusions, and four-contact enclosure bound for the zero-gap obstruction. It also introduces inner comparison points for the exact certificate, without changing the three frontier witnesses Q_-, Q_0, Q_+ .

E.1. Corner-clipped AB -unions. In this subsection the corner is the local origin O_{loc} , not the hexagon center O . The local anchors A, B used here are also distinct from the Newton comparison points introduced later. The corner chart

$$\Psi_4(u, v) = V_4 + u(V_5 - V_4) + v(V_3 - V_4)$$

has coordinate axes corresponding to the forward and backward edge directions. We identify a local vector $ue_A + ve_B$ with its pair (u, v) when applying this chart.

Let e_A, e_B be unit vectors with angle 120° , put

$$O_{\text{loc}} = 0, \quad A = ae_A, \quad B = be_B, \quad \mathcal{C}_{\text{loc}} = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$(60) \quad \mathcal{R}_{AB}^{\text{loc}}(a, b) = \mathcal{C}_{\text{loc}} \cap \bigcup \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ O_{\text{loc}}, A, B \in T \end{array} \right\}.$$

By the wedge reduction, the global union from the body is $\Psi_4(\mathcal{R}_{AB}^{\text{loc}}(b, a))$; its two anchor parameters are exchanged.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem E.1 (Strict supercritical AB -union). *Assume $a, b > 0$ and*

$$a + b > 1, \quad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\begin{aligned} \alpha &= \frac{h(a + 2b - aD)}{2\rho}, & \beta &= \frac{h(a - b + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-a + b + (a + b)D)}{2\rho}, & \delta &= \frac{h(2a + b - bD)}{2\rho}. \end{aligned}$$

Then all four coefficients are positive and

$$(61) \quad \mathcal{R}_{AB}^{\text{loc}}(a, b) = (\mathcal{C}_{\text{loc}} \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (\mathcal{C}_{\text{loc}} \cap \mathcal{D}_B \cap \mathcal{H}_B),$$

where

$$\mathcal{D}_A = \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\},$$

$$\mathcal{D}_B = \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\},$$

$$\mathcal{H}_A = \{(u, v) : \alpha(u - a) + \beta v \leq 0\},$$

$$\mathcal{H}_B = \{(u, v) : \gamma u + \delta(v - b) \leq 0\}.$$

Its non-axis frontier consists, in counterclockwise order, of the A -circle, the A -line, the B -line, and the B -circle.

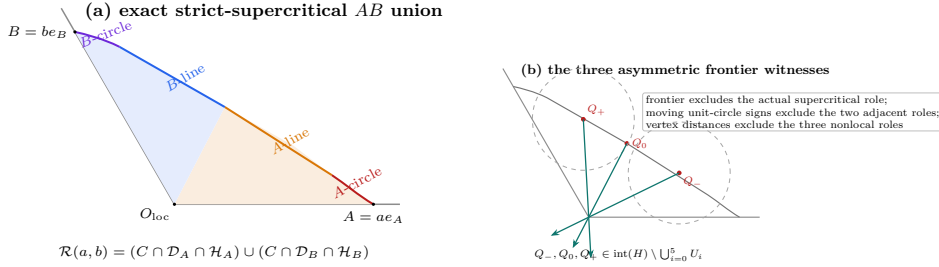


FIGURE 6. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A -circle arc, an A -line segment, a B -line segment, and a B -circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

Proof. The identities

$$4\rho - 3(a + b)^2 = (a - b)^2, \quad (a + b)^2 D^2 - (a - b)^2 = 4\rho((a + b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem 5.3. For a point $x = (u, v)$ in the relative interior of $\mathcal{C}_{\text{loc}}$ but outside $\triangle O_{\text{loc}}AB$, the four hull vertices occur in the order O_{loc}, B, x, A . The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h , are respectively

$$\max\{a, u\} + \max\{b, v - u\}, \quad \max\{b, v\} + \max\{a, u - v\},$$

and are at least $a + b > 1$. It remains to test the Ax and Bx calipers.

For the Ax caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

$$(62) \quad \frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \quad N_A = \begin{cases} 0, & p \leq 0, \\ ap, & 0 \leq p \leq v, \\ (a+b)p - bv, & p \geq v. \end{cases}$$

No support label is omitted in these three sectors. If $p \leq 0$, $G_A \leq h$ is equivalent to

$$r \leq 1, \quad -ap + (a+b)v \leq r.$$

The second inequality is the half-plane condition in (61), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$c = a + 2b - aD$, $d = a - b + (a+b)D$, $c' = a + 2b + aD$, $d' = a - b - (a+b)D$, and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \leq v$ one has $r \leq v$, whereas the numerator in (62) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \geq v$, the inequalities $G_A \leq h$ imply

$$(63) \quad r^2 + (a+b)p - bv \leq r, \quad bp + av \leq r.$$

Write $x - A = ry(\theta)$, $0 \leq \theta \leq 60^\circ$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}} \cos(\theta - 30^\circ), \quad w = \frac{2}{\sqrt{3}} \sin \theta.$$

Then (63) gives

$$r \leq f(\theta) = 1 - (a+b)t + bw, \quad S(\theta) = bt + aw \leq 1.$$

Let n_B be the unit vector with dual coordinates (γ, δ) . The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1+D), \quad b\delta - a\gamma = \frac{h}{2}(1-D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1-D)/2$. The function S has at most one critical point, a maximum, so $S(\theta) \leq 1$ implies $\theta \leq \theta_B$. On the feasible part $f, f' \geq 0$; hence $f(\theta) \cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B \rangle \leq a\gamma - b\delta + \frac{h}{2}(1-D) = 0.$$

The diameter bound also gives $\|x - B\| \leq 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges a, u with b, v and supplies the corresponding result for the Bx caliper. This proves (61) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line-circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2 + \alpha\beta + \beta^2 = h^2, \quad \gamma^2 + \gamma\delta + \delta^2 = h^2,$$

and the line determinant is

$$\alpha\delta - \gamma\beta = \frac{3(1-D)(D+3)}{16\rho} > 0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D)+2b}{2a+b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4-(1-D)^2) > 0$ and $b(4-(1-D)^2) > 0$. This proves the asserted four-piece order. $\square$

E.2. Exact-one handoffs and common reach bounds.

Lemma E.2 (Exact-one handoff chain). *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.12 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \quad b = \xi_4, \quad p = 1 - b, \quad q = 1 - a,$$

one has

$$(64) \quad 0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1,$$

and every selected V triangle obeys

$$a_i \geq p, \quad b_i \geq q.$$

Proof. Every selected V triangle other than 4 is strictly nonsupercritical, so $\xi_i < \xi_{i-1}$ for $i \neq 4$, which is precisely (E.2). In particular ξ_3 is the global minimum and ξ_4 the global maximum. The two anchors of V triangle 4 lie in its open triangle; their mutual squared distance is $a^2 + ab + b^2$, strictly below the diameter squared of the closed unit triangle. This proves (64). Finally every $\xi_j \in [\xi_3, \xi_4]$, so $1 - \xi_{i-1} \geq 1 - \xi_4 = p$ and $\xi_i \geq \xi_3 = q$. $\square$

E.3. Six directly forced radial points.

Put $h = \sqrt{3}/2$ and

$$s = p + q, \quad m = \min(p, q), \quad M = \max(p, q), \quad \chi = s^4 - s^2 + pq.$$

The strict domain (64) gives

$$(65) \quad pq > (1-s)(2-s), \quad m > 1-s, \quad s > 1-m.$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition A.7. On the present subcritical domain, equation (38) becomes

$$(66) \quad c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s .

Lemma E.3 (Direct radial forcing). *One has*

$$(67) \quad 0 < c_* < 1, \quad c_* > 1 - m.$$

Assume that $U_C, U_0, \dots, U_5$ cover H , with no restriction on the normalized V types. Then, for every i ,

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$(68) \quad \mathcal{D}_\eta = \{x : \|x\| \leq \eta\}, \quad \eta = h(1 - c_*).$$

Proof. The complementary pair is positive and has $p + q < 1$. With $m = \min(p, q)$ and $c_* = c_{\max}(p, q)$, lemma D.2 gives $1 - c_* \geq m(1 - m)/2 > 0$, while lemma 5.6 gives $1 - c_* < m$. Hence $0 < c_* < 1$ and $c_* > 1 - m$. Every selected pair dominates (p, q) by lemma E.2. Apply corollary 5.10 to obtain the unchanged six radial points and their inscribed disk. $\square$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = \Psi_4(u, v), \quad \mathbf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a . Accordingly, set

$$\begin{aligned} \alpha &= \frac{h(b + 2a - bD)}{2\rho}, & \beta &= \frac{h(b - a + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-b + a + (a + b)D)}{2\rho}, & \delta &= \frac{h(2b + a - aD)}{2\rho}, \end{aligned}$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The frontier coefficients here are not the signed-center slacks. Applying Theorem E.1 with the exchanged pair (b, a) , the two local line pieces are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \quad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D + 3))$ in

$$(69) \quad J = \left(\frac{2b + a - aD}{D + 3}, \frac{b + 2a - bD}{D + 3} \right).$$

Put

$$E_- = (1 - b, 2 - b), \quad E_+ = (2 - a, 1 - a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1 + t, t), \quad 1 - a \leq t \leq b, \quad \text{and} \quad Z_3(y) = (y, 1 + y), \quad 1 - b \leq y \leq a.$$

The restrictions of the two unit-circle equations are

$$(70) \quad \begin{aligned} g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \end{aligned}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned} \lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}. \end{aligned}$$

Lemma E.4 (Normalized two-coefficient bounds). *Suppose $x, y > 0$, $x^2 + xy + y^2 = 1$, $r > 1$, $0 < c < 1$, and $rx + cy = 1$. Then*

$$0 < x, y < 1, \quad cx < \frac{1}{\sqrt{3}}, \quad x - y < 2(1 - c).$$

Proof. Since $cy < 1 - x$ and $1 - x^2 = y(x + y)$,

$$c < \frac{1 - x}{y} = \frac{x + y}{1 + x}.$$

Consequently

$$cx < \frac{x(x + y)}{1 + x} \leq \frac{x + y}{2} \leq \frac{1}{\sqrt{3}}.$$

The last inequality follows from $1 = x^2 + xy + y^2 \geq 3(x + y)^2/4$. Also $1 - c > (1 - y)/(1 + x)$. If $x < y$, the second claimed bound is immediate; otherwise

$$x - y \leq 1 - y < (1 + x)(1 - c) < 2(1 - c).$$

□

Lemma E.5 (Moving adjacent-triangle signs). *Throughout (64),*

$$(71) \quad \lambda_* < 1/\sqrt{3} < \lambda_- < h^{-1}, \quad \mu_* < 1/\sqrt{3} < \mu_+ < h^{-1}.$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies the stronger bound

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 2 - a < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P, t) = \mathbf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1 - a, b]$,

$$(72) \quad F(L_+(\mu_+), t) \geq 0, \quad F(J, t) > 0, \quad F(L_-(\lambda_-), t) > 0.$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, the analogous signs hold for every $y \in [1 - b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Proof. Put $U = a + b - 1$, $z = a - b$, and normalize the line coefficients by

$$x = \alpha/h, \quad y = \beta/h, \quad w = \delta/h, \quad v = \gamma/h.$$

Direct substitution in their definitions gives

$$x^2 + xy + y^2 = w^2 + wv + v^2 = 1, \quad (a + b)x + by = (a + b)w + av = 1.$$

All four coefficients are positive. Also $0 < U < 2/\sqrt{3} - 1 < 1/6$ and $0 < D < 1$. Lemma E.4 therefore gives

$$(73) \quad bx < 1/\sqrt{3}, \quad aw < 1/\sqrt{3}, \quad x - y < 2(1 - b), \quad w - v < 2(1 - a).$$

Reparametrize the two frontier lines by $k = h\lambda$ and $k = h\mu$:

$$P_-(k) = (b - yk, xk), \quad P_+(k) = (wk, a - vk).$$

Their common junction parameter is

$$k_* := h\lambda_* = h\mu_* = \frac{2\rho}{D + 3} = \frac{D^2 + 3}{2(D + 3)}, \quad \frac{3}{8} < k_* < \frac{1}{2}.$$

The circle restrictions become

$$f_b(k) := \mathbf{q}(P_-(k) - E_-) - 1 = k^2 - 3(1 - Ux)k + 3b^2 - 3b + 2,$$

$$f_a(k) := \mathbf{q}(P_+(k) - E_+) - 1 = k^2 - 3(1 - Uw)k + 3a^2 - 3a + 2.$$

For the first quadratic,

$$f_b(1/2) = 3(b - 1/2)^2 + \frac{3}{2}Ux > 0, \quad f'_b(1/2) = -2 + 3Ux < -\frac{3}{2}.$$

Moreover

$$f_b(1) = 3\{Ux - b(1 - b)\} < 0,$$

since $\rho = 1 + U(1 + a) - b(1 - b) < 1$ and $x < 1 < 1 + a$. Thus its first root $k_- = h\lambda_-$ lies in $(1/2, 1)$. The reflected calculation gives $k_+ = h\mu_+ \in (1/2, 1)$. Both quadratics decrease on $[k_*, 1/2]$, so

$$f_b(k_*) > f_b(1/2) > 0, \quad f_a(k_*) > f_a(1/2) > 0.$$

This proves the root ordering and the positive junction values without expanding a junction polynomial.

For the junction coordinates, use

$$D^2 - (3U - z)^2 = 6U(1 + z - U) > 0, \quad D^2 - (3U + z)^2 = 6U(1 - z - U) > 0.$$

The first identity implies $3U - z < D(1 + 2a)$, which is exactly $J_u < 1/2$; the second gives $J_v < 1/2$. Hence $J_u + J_v < 1$. Because

$$\partial_t F(P, t) = 1 + 2t - P_u - P_v,$$

$F(J, t)$ is strictly increasing for $t \geq 0$. Its value at $t = 1 - a$ is $f_a(k_*) > 0$, proving the junction sign.

On the opposite line, direct differentiation gives

$$\frac{d}{dk} F(P_-(k), t) = 2k + (1 - b)(2y + x) + t(y - x).$$

For $k \geq k_*$ and $t \in [1 - a, b]$, this is at least

$$2k - bx > \frac{3}{4} - \frac{1}{\sqrt{3}} > 0.$$

Thus $F(P_-(k_-), t) > F(J, t) > 0$ throughout the handoff interval.

On the correct line, $0 < k_+ < 1$ and (73) give

$$(P_+(k_+))_u + (P_+(k_+))_v = a + k_+(w - v) < 2 - a.$$

Indeed, if $w - v < 0$ the sum is below $a < 2 - a$; otherwise it is below $a + 2(1 - a)$. Therefore, for $t \geq 1 - a$,

$$\partial_t F(P_+(k_+), t) > 1 + 2(1 - a) - (2 - a) = 1 - a > 0.$$

As $F(P_+(k_+), 1 - a) = f_a(k_+) = 0$, its sign is nonnegative throughout the interval and positive away from its left endpoint. Reflection proves all the remaining assertions. $\square$

Define the local and global witnesses by

$$(74) \quad Q_-^{\text{loc}} = L_-(\lambda_-), \quad Q_0^{\text{loc}} = J, \quad Q_+^{\text{loc}} = L_+(\mu_+),$$

and put $Q_j = \Psi_4(Q_j^{\text{loc}})$ for $j \in \{-, 0, +\}$.

Lemma E.6 (Direct asymmetric forcing). *Assume that $U_C, U_0, \dots, U_5$ cover H . Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i,$$

and consequently all three witnesses belong to U_C .

Proof. Put $\bar{\alpha} = \alpha/h$, $\bar{\beta} = \beta/h$, and $\bar{\gamma} = \gamma/h$. Write $x = \bar{\alpha}$, $y = \bar{\beta}$. The coefficient identities give $\rho - 1 = (y - b)(x + y - b)/x^2$. Since $x + y > 1 > b$, one has $y < b$; reflection gives $\bar{\gamma} < a$. Thus both outer line endpoints at $k = 1$ have coordinates in $(0, 1)^2$. The junction and the two first-root points lie on the intervening open segments, so $0 < u, v < 1$ for all three points. Each point has local coordinates

in $\mathcal{R}_{AB}^{\text{loc}}(b, a)$ and hence lies in a closed unit triangle containing V_4 , so $\mathbf{q}(u, v) \leq 1$; hence $|u - v| < 1$ and all three points lie in $\text{int}(H)$.

In local coordinates the closure of U_4 is represented in $\mathcal{R}_{AB}^{\text{loc}}(b, a)$, since U_4 contains the actual handoffs $X_3(\xi_3), X_4(\xi_4)$ and its distinguished vertex V_4 . All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle U_4 , a sufficiently small plane ball about it would lie both in U_4 and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}_{AB}^{\text{loc}}(b, a)$. This contradiction excludes U_4 .

For the triangles U_0, U_1, U_2 , their contained vertices have local coordinates

$$\begin{array}{c|ccc} i & 0 & 1 & 2 \\ \hline V_i & (2, 1) & (2, 2) & (1, 2). \end{array}$$

For $P = (u, v)$,

$$\|P - (2, 1)\|^2 - 1 = \mathbf{q}(u, v) + 2 - 3u, \quad \|P - (1, 2)\|^2 - 1 = \mathbf{q}(u, v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from V_0 for Q_-, Q_0 and from V_2 for Q_0, Q_+ . For the remaining pair, the first-root circle and the sum bound in Lemma E.5 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2, 2)$, because writing $u = 1 - \xi$, $v = 1 - \eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude U_0, U_1, U_2 .

The actual handoff $X_5(\xi_5) \in U_5$ has local coordinates $Z_5(\xi_5)$, and $1 - a < \xi_5 < b$ by (E.2). The three signs in (72) therefore put every witness at distance at least one from $X_5(\xi_5)$, excluding U_5 . Similarly, if $y = 1 - \xi_2$, then $1 - b < y < a$ and $X_2(\xi_2) \in U_3$ has local coordinates $Z_3(y)$; the reflected signs exclude U_3 . Thus no V triangle contains any witness. The assumed cover now forces all three interior witnesses into U_C . $\square$

Together with lemma E.3, this proves the containment $K_{\text{wit}}(a, b) \subset U_C$ used in the body.

E.4. Fixed-start Newton inner reduction. The three frontier witnesses have already been forced into U_C . We move the outer two toward the junction only for the enclosure comparison. These inner points belong to U_C by convexity; they need not be missed by every V triangle.

Write $U = a + b - 1$ and normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients. For $k = h\lambda$ and $k = h\mu$, the circle restrictions are

$$\begin{aligned} f_b(k) &= k^2 - 3(1 - U\bar{\alpha})k + 3b^2 - 3b + 2, \\ f_a(k) &= k^2 - 3(1 - U\bar{\delta})k + 3a^2 - 3a + 2. \end{aligned}$$

The preceding root bounds place $1/2$ strictly between the junction parameter and either first root. A Newton step from this fixed value gives

$$\hat{k}_b = \frac{1 + 3(b - 1/2)^2}{2 - 3U\bar{\alpha}}, \quad \hat{k}_a = \frac{1 + 3(a - 1/2)^2}{2 - 3U\bar{\delta}}.$$

The denominators exceed $3/2$. To keep the line-parameter notation set $\widehat{\xi}_- = 4\widehat{k}_b/3$ and $\widehat{\zeta}_+ = 4\widehat{k}_a/3$. Define the global points

$$\begin{aligned} A &= \Psi_4(b - \bar{\beta}\widehat{k}_b, \bar{\alpha}\widehat{k}_b), \\ B &= \Psi_4(J), \\ C &= \Psi_4(\bar{\delta}\widehat{k}_a, a - \bar{\gamma}\widehat{k}_a). \end{aligned}$$

All subsequent norms and determinants refer to these global vectors.

Lemma E.7 (Fixed-start Newton inner reduction). *The local coordinates are rational in a, b, D , and the global coordinates are rational over $\mathbb{Q}(\sqrt{3})$ in those variables. One has*

$$A \in (Q_0, Q_-), \quad C \in (Q_0, Q_+).$$

Consequently, for the actual disk $\mathcal{D}_\eta$,

$$\widehat{K} = \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\})$$

satisfies

$$(75) \quad \widehat{K} \subseteq \text{conv}(K_{\text{wit}}(a, b)), \quad \Lambda(K_{\text{wit}}(a, b)) \geq \Lambda(\widehat{K}).$$

Proof. Each quadratic is positive and decreasing at $k = 1/2$, and has positive second derivative. Its tangent zero is therefore strictly between $1/2$ and its first root. The junction is before $1/2$ by lemma E.5. This proves both open-segment inclusions. The displayed formulas are rational in the stated fields, and convexity proves the set inclusion. $\square$

These points lie farther along the two segments than the earlier junction-start Newton points: for a positive decreasing convex quadratic, the Newton map has derivative $ff''/(f')^2 > 0$ before the first root. The four-contact proof below uses only the segment inclusions and the unchanged supporting lines; its geometric reasoning is therefore unchanged. The old algebraic certificate is not reused for the new point formulas.

E.5. Ordered convex chain and supporting-line bounds.

Proposition E.8 (Four-contact witness geometry). *Assume (64) and $c_* > 2/3$. The rays A, B, C occur in that order in one open 120° cone, and*

$$\text{cross}(B - A, C - B) > 0, \quad \|A\|, \|C\| > \eta.$$

Moreover the outward distances of the lines AB, BC satisfy

$$(76) \quad d_{AB} := \frac{\text{cross}(A, B)}{\|B - A\|} \geq h - 2\eta, \quad d_{BC} := \frac{\text{cross}(B, C)}{\|C - B\|} \geq h - 2\eta.$$

Proof. The local coordinates of all three Newton points lie in $(0, 1)^2$. Centering at O gives coefficient vectors $(u - 1, v - 1)$ in the two incident edge directions, so their rays lie in one open 120° cone. The two line identities

$$d_{AB} = \alpha(1 - b) + \beta > 0, \quad d_{BC} = \gamma + \delta(1 - a) > 0$$

give their order A, B, C . Put $s_A = h\widehat{\xi}_- - \lambda_* > 0$ and $s_C = h\widehat{\zeta}_+ - \mu_* > 0$. The frontier directions give

$$\text{cross}(B - A, C - B) = hs_A s_C (\alpha\delta - \beta\gamma) > 0,$$

since $\alpha\delta - \beta\gamma = 3(1-D)(D+3)/(16\rho) > 0$. For $A = (-X, -Y)$ in the centered cone basis, $X > 1/2$, and

$$\|A\|^2 = (Y - X/2)^2 + 3X^2/4 > 3/16.$$

The reflected calculation gives

$$(77) \quad \|A\|, \|C\| > h/2 > \eta.$$

It remains to prove the two supporting-line bounds. Put $\tau = h - 2\eta = h(2c_* - 1)$. We now give the exact analytic verification. By reflection take $a \leq b$ and set $m = 1 - b$, $M = 1 - a$, $U = a + b - 1$, $s = m + M$, and $w = M - m$. The smaller of the two normalized left sides in (76) is

$$(78) \quad d = \frac{\mathcal{A} + U(2m - 1) + D(\mathcal{A} + U)}{2\rho}, \quad \mathcal{A} = 1 - m + m^2.$$

Apply lemma D.2 to the complementary pair (m, M) . Its shared deficit bound gives $2c_* - 1 \leq 1 - m + m^2 = \mathcal{A}$. On the C_L sheet, put

$$X_0 = \mathcal{A} + U, \quad Y_0 = 2\rho\mathcal{A} - \mathcal{A} - U(2m - 1).$$

Then

$$d - \mathcal{A} = \frac{D(\mathcal{A} + U) - Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned} Y_0 &= 2(m^2 - m + 1)U^2 + (2m^3 - 2m + 3)U \\ &\quad + (m^2 - m + 1)(2m^2 - 2m + 1) > 0. \end{aligned}$$

Furthermore

$$D^2 X_0^2 - Y_0^2 = 4m\rho\Phi(U, m),$$

where

$$\begin{aligned} \Phi(U, m) &= (1 - m)(m^2 - m + 2)U^2 \\ &\quad + (2 - m^2)(m^2 - m + 1)U \\ &\quad - m(1 - m)^2(m^2 - m + 1). \end{aligned}$$

In these variables

$$\chi = U^4 - 4U^3 + 5U^2 - (m + 2)U + m(1 - m).$$

Since $U^2(U^2 - 4U + 5) > 0$, the selector $\chi \leq 0$ implies $U > m(1 - m)/(m + 2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1 - m)(m^4 - 2m^3 + 6m^2 - 6m + 4)}{(m + 2)^2} > 0.$$

The last quartic equals $(m^2 - m)^2 + 5m^2 - 6m + 4 > 0$. Thus $d > \mathcal{A} \geq 2c_* - 1$.

On the other radial sheet let $E = \sqrt{4s^2 - 3}$. Then $0 \leq w < sE$ and $c_* = (s + w)/(1 + E)$. At fixed U , differentiation of (78) can be checked without an implicit estimate. Namely,

$$d = \frac{1 + D}{2} - UB(D, w), \quad B(D, w) = \frac{(1 + U)(3 + D) + w(1 - D)}{D^2 + 3}.$$

Here $D^2 = w^2 + 6U + 3U^2$, so $0 \leq D' = w/D \leq 1$, and

$$B_w = \frac{1 - D}{D^2 + 3} \geq 0.$$

Writing $N = (1 + U)(3 + D) + w(1 - D)$, direct differentiation gives

$$B_D = \frac{(1 + U - w)(D^2 + 3) - 2DN}{(D^2 + 3)^2} > -\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1 + U - w = 2a > 0$, $N < (7/6)4 + 1 = 17/3$, $D < 1$, and $D^2 + 3 \geq 3$. If $B_D < 0$, then $B_D D' \geq B_D$; otherwise $B_D D' \geq 0$. Hence $B' = B_w + B_D D' > -34/27$, and $U < 1/6$ yields

$$d' < \frac{1}{2} + \frac{1}{6} \frac{34}{27} = \frac{115}{162} < 1,$$

whereas $(2c_* - 1)' = 2/(1 + E) \geq 1$. Hence their difference decreases with w . At the boundary $w = sE$, where $\chi = 0$ and $c_* = s$, the preceding sheet applies: this is an admissible point because $h < s < 1$ and $D^2 = 4s^4 - 12s + 9 < 1$. Indeed $s^4 - 3s + 2 = (s - 1)(s^3 + s^2 + s - 2) < 0$ on this interval; the second factor is increasing and already equals $(7h - 5)/4 > 0$ at $s = h$. The preceding sheet gives $d \geq 2s - 1$. Therefore (76) holds on both sheets, and reflection supplies the other line.

This proves both supporting-line bounds. Since $0 < \eta < h/3$, their common lower bound $h - 2\eta$ is strictly greater than η , as required for the exposed-edge geometry used below. $\square$

E.6. Four supporting contacts and paired radius transfer. Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For the inner witness define

$$H(n) = \max\{\eta, \langle A, n \rangle, \langle B, n \rangle, \langle C, n \rangle\}, \quad \Psi(n) = \sum_{j=0}^2 H(R^j n).$$

The support formula gives $\Lambda(\widehat{K}) = h^{-1} \min_{\|n\|=1} \Psi(n)$. The minimum can be located without introducing another convex body.

Lemma E.9 (Four-contact enclosure formula). *For the Newton inner witness in the range $c_* > 2/3$, put*

$$\begin{aligned} n_{AB} &= -\frac{J(B - A)}{\|B - A\|}, & n_{BC} &= -\frac{J(C - B)}{\|C - B\|}, \\ t_A &= \frac{\eta A - \sqrt{\|A\|^2 - \eta^2} JA}{\|A\|^2}, & t_C &= \frac{\eta C + \sqrt{\|C\|^2 - \eta^2} JC}{\|C\|^2}. \end{aligned}$$

Then

$$(79) \quad \Lambda(\widehat{K}) = \frac{1}{h} \min\{\Psi(n_{AB}), \Psi(n_{BC}), \Psi(t_A), \Psi(t_C)\}.$$

Proof. By proposition E.8, the rays A, B, C are ordered in one open 120° cone, the chain turns strictly counterclockwise, and the supporting-line distances exceed η . The third point and the disk are strictly on the inward side of each of AB, BC , so both segments are exposed edges. Moreover

$$\text{cross}(C - A, B - A) < 0, \quad \text{cross}(C - A, -A) > 0.$$

Thus B and the origin are on opposite sides of AC , which is not an exposed edge. The other two straight boundary pieces are the exposed disk tangents through A and C . There is no exposed disk tangent through B . The indicated signs select

the tangent before A and the tangent after C in the counterclockwise boundary traversal.

The support-cell rotation lemma lemma 5.2 places a minimum at these exposed ties or on a disk-only cell. In the latter case the same value occurs at a boundary tie, since $\|A\| > \eta$. Rotation through 120° leaves just the four displayed normals. The exposed-hull argument, rather than the general caliper theorem alone, is what discards the other point and disk ties. $\square$

At the first two contacts the disk supplies the other two supports, so

$$(80) \quad \Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h, \quad \Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h.$$

For the two tangencies, put

$$\Delta = \text{cross}(C, RA) > 0, \quad \nu = \langle C, RA \rangle, \quad N_X = \|X\|^2 \quad (X = A, C).$$

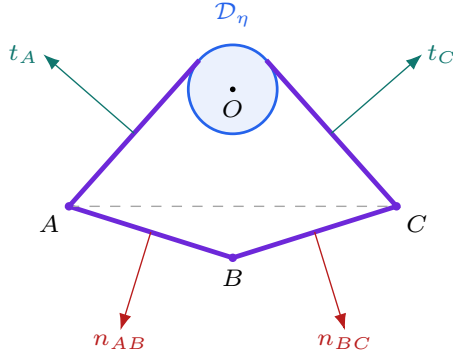
Choosing C as a support source at Rt_A and A at R^2t_C , with the disk in the other directions, gives

$$(81) \quad \Psi(t_A) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_A - \eta^2}}{N_A}, \quad \Psi(t_C) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_C - \eta^2}}{N_C}.$$

Consequently the only remaining targets are the paired residual signs

$$(82) \quad P_X(\eta) := (N_X - \eta^2)\Delta^2 - ((h - 2\eta)N_X - \eta\nu)^2 \geq 0 \quad (X = A, C).$$

(a) the four exposed contacts



(b) four support sums

Two consecutive-point contacts:

$$\Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h,$$

$$\Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h.$$

Two exposed disk tangencies:

$$P_A(\eta), P_C(\eta) \geq 0$$

$$\implies \Psi(t_A), \Psi(t_C) \geq h.$$

$$\text{Hence } \Lambda(\hat{K}) \geq 1.$$

FIGURE 7. The four-contact terminal for the zero-gap witness. The supporting lines through AB and BC and the two outer disk tangencies exhaust the active changes of support source. The dashed segment AC is not exposed: B and the disk center lie on opposite sides. The drawing is schematic; the exact witness geometry and the paired tangent residuals are proved in the text. No auxiliary Minkowski sum or cap covering is needed.

Indeed $P_X(\eta) \geq 0$ gives $\Delta\sqrt{N_X - \eta^2} \geq |(h - 2\eta)N_X - \eta\nu|$, which implies the required lower bound even when the expression inside the absolute value is negative. No sign is lost by squaring.

The exact certificate uses a smaller rationally specified radius. The following elementary transfer is the reason those existing signs suffice for the actual-radius tangencies.

Lemma E.10 (Simultaneous radius transfer). *Let X, Y be vectors, $d \geq 0$, and $0 \leq e_0 \leq 1/3$. If*

$$\|(1 - 2e_0)X - e_0Y\| \leq d, \quad \|(1 - 2e_0)Y - e_0X\| \leq d,$$

then both inequalities hold at every $e \in [e_0, 1/3]$.

Proof. The difference of the two vectors at e_0 is $(1 - e_0)(X - Y)$, so

$$\|X - Y\| \leq \frac{2d}{1 - e_0} \leq 3d.$$

At $e = 1/3$ the vectors become $(X - Y)/3$ and $(Y - X)/3$, both in the closed radius- d ball. Each vector depends affinely on e , and the ball is convex. Interpolation proves both inequalities, including the endpoint case. This transfers the pair together; it does not assert monotonicity of either squared residual separately. $\square$

Lemma E.11 (Solved witness-enclosure optimization). *For every parameter pair in equation (20),*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Proof. If $c_* \leq 2/3$, the centered disk gives $\Lambda(K_{\text{wit}}) \geq 3(1 - c_*) \geq 1$. Otherwise $0 < \eta < h/3$, and it suffices to enclose $\hat{K}$ by equation (75). The line contacts satisfy equation (80).

theorem F.6 proves the paired residual signs for the fixed-start points at $0 < e_0 \leq 1 - c_*$. The Gram identities and actual-radius transfer in lemma F.3 give $P_A(\eta), P_C(\eta) \geq 0$. Thus both tangent sums in equation (81) are at least h as well. All four contacts in equation (79) have support sum at least h . Therefore

$$\Lambda(K_{\text{wit}}) \geq \Lambda(\hat{K}) \geq 1.$$

$\square$

APPENDIX F. EXPLICIT DISK-TANGENT COMPARISONS

The frontier witnesses remain Q_-, Q_0, Q_+ . The comparison points A, B, C are the fixed-start Newton points of lemma E.7; they are not additional forced frontier witnesses. The actual disk has radius $\eta = h(1 - c_*)$. Only the tangent-residual calculation uses a smaller scalar radius. The construction below replaces the former sparse Bernstein transcript, not the geometric forcing or the two straight-line contact estimates. It remains an exact computer-assisted argument. The five new comparison checks use 383 Chebyshev coefficients, the curvature comparison uses 113, and the two increment comparisons use 208 nonconstant coefficients; the boundary Taylor and identity checks are additional. The historical junction-start transcript and its two exact programs remain unchanged regression material, not inputs for these new point formulas.

F.1. A single rational radius envelope. Reflect so that $a \geq b$ and set

$$\begin{aligned} m &= 1 - a, & U &= a + b - 1, & a &= 1 - m, & b &= m + U, \\ \omega &= m(1 - m), & b_0 &= \omega/2. \end{aligned}$$

Here U is a scalar, not an open triangle U_i . The quartic capacity root $C_L(m)$ was defined in equation (66). We first obtain a rational lower bound for $1 - C_L(m)$, then extend it to both capacity branches.

Lemma F.1 (A rational upper-start Newton deficit). *Let $0 < m < 1/2$, $h = \sqrt{3}/2$, and let $C_L(m)$ be the unique root in $[h, 1]$ of $F_m(c) = c^4 - c^2 + mc - m^2$. Set $\omega = m(1 - m)$ and $c_0 = 1 - \omega/2$. Then*

$$r(m) := 1 - c_0 + \frac{F_m(c_0)}{F'_m(c_0)} = \frac{\omega(4 - \omega)(3\omega^2 - 4\omega + 4)}{8(4 + 2m - 10\omega + 6\omega^2 - \omega^3)}$$

satisfies

$$0 < \omega/2 < r(m) \leq 1 - C_L(m) < m.$$

Proof. One has $c_0 \geq 7/8 > h$ and

$$F_m(c_0) = \frac{m^2(1 - m)}{16}P(m), \quad P(m) = 12 - 28m + 17m^2 - 11m^3 + 3m^4 - m^5.$$

On $[0, 1/2]$, $P' \leq -28 + 17 + 3/2 = -19/2$ and $P(1/2) = 33/32 > 0$. Thus $F_m(c_0) > 0$. The polynomial F_m is increasing and strictly convex on $[h, 1]$, so $c_0 > C_L(m)$ and its tangent zero $c_1 = c_0 - F_m(c_0)/F'_m(c_0)$ satisfies $C_L(m) \leq c_1 < c_0$. This proves the first bounds. If $1 - m < h$, then $C_L(m) > 1 - m$ immediately. Otherwise it follows from

$$F_m(1 - m) = -m((1 - m)^3 + m^2) < 0.$$

The displayed rational formula follows by substitution; its denominator factor $4 + 2m - 10\omega + 6\omega^2 - \omega^3 = 2F'_m(c_0)$ is positive. $\square$

Lemma F.2 (Uniform affine radius envelope). *In the strict F domain, put $U = a + b - 1$, $m = \min(1 - a, 1 - b)$, and let c_* be the exact common-pair capacity. For any $0 < r \leq 1 - C_L(m)$, define*

$$E_r(m, U) = r + (m - r) \left(1 - \frac{U}{r}\right)_+ = \max \left\{ r, m - \frac{m - r}{r}U \right\}.$$

Then

$$1 - c_* \geq E_r(m, U).$$

In particular, the rational function in Lemma F.1 gives one explicit radius recipe without testing the original capacity-branch selector.

Proof. Reflect to $a \geq b$ and set $s = 1 - U$, $C = C_L(m)$. The exact capacity formula is $c_* = C$ on the quartic branch and

$$g_m(s) = \frac{2(s - m)}{1 + \sqrt{4s^2 - 3}}$$

on the triangular branch. On the latter, its selector $\chi = F_m(s) > 0$ gives $s > h$ because $\chi \leq s^4 - 3s^2/4$, and hence $s > C$. Writing $E = \sqrt{4s^2 - 3}$,

$$g'_m(s) = \frac{2(E - 3 + 4ms)}{E(1 + E)^2} \leq 0 \quad (C < s \leq 1).$$

The root equation gives $m/C = (1 \pm \sqrt{4C^2 - 3})/2$, and substitution shows $g_m(C) \leq C$. Thus $c_* \leq C$ on both branches, so $1 - c_* \geq r$.

It remains to consider $0 < U < r$. Here $s > 1 - r \geq C$, so the triangular formula applies. On the analytic interval $s \in [1 - r, 1]$,

$$g_m''(s) = \frac{8\{s(3 + 9E - 2E^2) - m(3 + 9E + 2E^3)\}}{E^3(1 + E)^3} > 0.$$

Indeed $m/s \leq 1/\sqrt{3} < 2/3$ and

$$(3 + 9E - 2E^2) - \frac{2}{3}(3 + 9E + 2E^3) = 1 + 3E - 2E^2 - \frac{4}{3}E^3 \geq \frac{2}{3} \quad (0 \leq E \leq 1),$$

since the last cubic is concave and lies above its endpoint chord. Endpoint cases follow by continuity. Therefore $U \mapsto g_m(1 - U)$ is convex. Its endpoint bounds are $g_m(1) = 1 - m$ and $g_m(1 - r) \leq C \leq 1 - r$. The chord inequality gives

$$c_* \leq (1 - U/r)(1 - m) + (U/r)(1 - r),$$

which is the claimed affine bound. Combining it with $1 - c_* \geq r$ proves the result. $\square$

For this choice of r put

$$(83) \quad \kappa = \frac{m - r}{r}, \quad e_0 = \max\{r, m - \kappa U\}, \quad e_* = 1 - c_*.$$

Thus $0 < b_0 \leq r \leq e_0 \leq e_*$, $r < m$, and $r \leq 1 - h < 1/6$. The slope satisfies

$$(84) \quad 1 + \frac{m}{2} + 2m^2 \leq \kappa \leq 3, \quad \ell(m) := \frac{1 + 5m}{2} \leq \kappa.$$

For the upper bound use $r \geq m(1 - m)/2$. For the lower bound write $r = N_r/L_r$, using the positive numerator and denominator in lemma F.1. Exact expansion gives

$$mL_r - N_r - (1 + m/2 + 2m^2)N_r = \frac{m^3}{2}H_7(m),$$

where

$$H_7(m) = 4 + m(164 - 249m) + m^3(312 - 238m) + m^5(136 - 45m) + 12m^7 > 0.$$

Finally $1 + m/2 + 2m^2 - \ell(m) = 2(m - 1/2)^2$.

F.2. One residual and its exact polynomial definition. Let $X = A$, $Y = R^{-1}C$, and set

$$N_X = \|X\|^2, \quad N_Y = \|Y\|^2, \quad \nu = \langle X, Y \rangle, \quad \Delta_0 = \frac{\text{cross}(Y, X)}{h}.$$

The ordered-chain geometry makes $\Delta_0 > 0$. The Gram identity is

$$(85) \quad h^2 \Delta_0^2 = N_X N_Y - \nu^2.$$

For $t = 1 - 2e$ define

$$(86) \quad Q_X(e) = \Delta_0^2 - \|tX - eY\|^2, \quad Q_Y(e) = \Delta_0^2 - \|tY - eX\|^2.$$

These are tangent residuals, not the frontier points Q_j .

Use barred coefficients $\bar{\alpha} = \alpha/h$, and similarly for the other three frontier coefficients. Put

$$\begin{aligned} A_1 &= \rho\bar{\alpha}, & B_1 &= \rho\bar{\beta}, & D_1 &= \rho\bar{\delta}, & G_1 &= \rho\bar{\gamma}, & \rho &= a^2 + ab + b^2, \\ n_b &= 12b^2 - 12b + 7, & n_a &= 12a^2 - 12a + 7, & E_b &= 8\rho - 12UA_1, & E_a &= 8\rho - 12UD_1, \end{aligned}$$

$$\begin{aligned} J_b &= (1 - 2b)E_b + (A_1 + 2B_1)n_b, & L_b &= E_b - A_1n_b, \\ J_a &= (1 - 2a)E_a + (D_1 + 2G_1)n_a, & L_a &= E_a - D_1n_a. \end{aligned}$$

The denominators satisfy $E_a, E_b > 6\rho > 0$. In an orthonormal frame the two vectors are

$$X = (-J_b/(2E_b), -hL_b/E_b), \quad Y = (-J_a/(2E_a), hL_a/E_a).$$

Consequently $4E_a^2E_b^2Q_X(e)$ equals

$$\begin{aligned} \mathcal{N}(D, e) &= (J_bL_a + J_aL_b)^2 - ((1 - 2e)J_bE_a - eJ_aE_b)^2 \\ &\quad - 3((1 - 2e)L_bE_a + eL_aE_b)^2. \end{aligned}$$

Reducing modulo $D^2 = K := 4\rho - 3$ defines, by exact polynomial division,

$$(87) \quad \text{rem } \mathcal{N} = 2\rho^2(\mathcal{A} + D\mathcal{B}), \quad \Pi = (1 + 3K)\mathcal{A} + K(3 + K)\mathcal{B}.$$

Both $\mathcal{B}$ and Π are quadratic in e . Thus

$$Q_X(e) = \frac{\rho^2}{2E_a^2E_b^2}(\mathcal{A} + D\mathcal{B}).$$

No independently stored coefficient table defines these polynomials.

Lemma F.3 (Radius transfer to the actual tangencies). *If $0 < e_0 \leq e_* < 1/3$ and $Q_X(e_0), Q_Y(e_0) \geq 0$, then the two actual-radius tangent contact sums in equation (81) are at least h .*

Proof. Apply lemma E.10 to X, Y, Δ_0 . This gives $Q_X(e_*), Q_Y(e_*) \geq 0$. The Gram identity gives

$$(88) \quad P_A(he) = h^2 N_X Q_X(e).$$

$$(89) \quad P_C(he) = h^2 N_Y Q_Y(e).$$

The positive determinant sign and the unsquared implication after equation (82) give the two support bounds. Only the residual is tested at e_0 ; the disk in the line-contact and exposed-hull arguments remains the actual disk of radius he_* . $\square$

F.3. Explicit low-degree comparisons. For $s, t \in [-1, 1]$ write $m = (1 + s)/4$, $U = (1 + t)/12$. The elementary Chebyshev recurrence

$$T_0(z) = 1, \quad T_1(z) = z, \quad T_{n+1}(z) = 2zT_n(z) - T_{n-1}(z)$$

gives $|T_n(z)| \leq 1$ on $[-1, 1]$. If $F - g = \sum c_{ij}T_i(s)T_j(t)$, then

$$(90) \quad |F - g| \leq \sum_{i,j} |c_{ij}|.$$

The coefficients are computed over $\mathbb{Q}$, using the exact power conversion and then checking reconstruction. Thus each displayed error budget below is a finite rational inequality, not a sampling claim.

Lemma F.4 (Explicit algebraic comparison bounds). *On $\mathcal{R} = [0, 1/2] \times [0, 1/6]$ the polynomials defined by equation (87) satisfy*

$$\mathcal{B}(m, U, b_0) > 4, \quad \mathcal{B}(m, U, m) > 210, \quad \mathcal{B}_{ee} < -5600.$$

Put

$$\begin{aligned} H_1 &= -\Pi_{Uee}, & H_2 &= \Pi_{UUe}(m, U, 0) - 2\Pi_{Uee}, \\ H_3 &= \Pi_{UUe}(m, U, 1/3) - 2\Pi_{Uee}, & H_4 &= \Pi_{Ue}(m, U, 1/3). \end{aligned}$$

Then $H_1 > 250000$, $H_2 > 500000$, $H_3 > 550000$, and $H_4 > 20000$. Moreover

$$-\Pi_{UU}(m, U, 1/3) + 2\ell\Pi_{Ue}(m, U, 1/3) - \ell^2\Pi_{ee} > 21500.$$

Writing $x = 1 - 2m$, one also has

$$\begin{aligned}\Pi_U(m, U, b_0) &\geq 140 + 503x + \frac{6265}{2}x^2 \\ &\quad + 25000(U - b_0) \quad (U \geq b_0),\end{aligned}$$

$$\Pi_U(m, U, 1/3) > 10700 \quad (m \geq 1/3).$$

On the strict physical domain, $N_X \geq N_Y$ and $\Pi_{ee} < 0$. The latter sign also holds on $0 < U \leq r(m)$.

Proof. We specify the certificate rather than an existence assertion for a sum of squares. The four normalized targets $H_i/1000$ have the quadratic models

$$\begin{aligned}G_1 &= 130s^2 + 80st - 124s + 52t^2 + 100t + 454, \\ G_2 &= 466s^2 + 414st + 124s + 238t^2 + 771t + 1379, \\ G_3 &= 286s^2 + 228st - 202s + 154t^2 + 348t + 1090, \\ G_4 &= 18s^2 + 4st - 14s + 6t^2 + 17t + 44.\end{aligned}$$

Their exact Chebyshev error sums are bounded, respectively, by 70, 300, 176, 8. For $G = as^2 + bst + cs + dt^2 + jt + k$ complete the square as

$$G = a \left(s + \frac{bt + c}{2a} \right)^2 + \gamma(t + 1)^2 + \delta(t + 1) + L,$$

where

$$\gamma = d - \frac{b^2}{4a}, \quad \delta = j - \frac{bc}{2a} - 2\gamma, \quad L = k - j + d - \frac{(c - b)^2}{4a}.$$

For the four models $\gamma, \delta > 0$, and

$$L = \frac{21188}{65}, \quad \frac{373211}{466}, \quad \frac{210031}{286}, \quad \frac{57}{2},$$

respectively. Subtracting the error budgets proves the four bounds.

For $\mathcal{B}(m, U, b_0)$ use the cubic

$$\begin{aligned}10g &= -292s^3 + 556s^2t + 1504s^2 + 364st^2 - 317st - 1795s \\ &\quad - 28t^3 + 272t^2 - 389t + 804.\end{aligned}$$

Its exact error is less than 17/2. On $s \leq 0$, $g_{ss} > 0$ and $g_s(0, t) < 0$, so $g(s, t) \geq g(0, t) \geq 387/10$. On $0 \leq s \leq 1$, the affine matrix $10(\nabla^2 g - 8I)$ has positive upper-left entry and determinant at all four corners. Its determinant values are 55687, 1026919, 82551, 23383. Convexity of the positive-semidefinite cone gives $\nabla^2 g \succeq 8I$. At $(2/3, 1/3)$ the value is 383/30 and gradient is $(13/45, -35/18)$. Completing a square gives $g \geq 1623259/129600 > 25/2$, hence $\mathcal{B}(m, U, b_0) > 4$. The other two $\mathcal{B}$ bounds follow by exact coefficient grouping in $x = 1 - 2m \in [0, 1]$, $6U \in [0, 1]$.

For the curvature target divided by 100, use

$$\begin{aligned}g_c &= 76s^3 - 160s^2t - 184s^2 + 104st^2 + 226st - 132s \\ &\quad - 8t^3 - 34t^2 + 283t + 788.\end{aligned}$$

The exact Chebyshev error is less than 55. Setting $X_0 = 1 - s$, $Y_0 = 1 + t$ and grouping nonnegative products on $[0, 2]^2$ gives

$$g_c \geq 277 - 37X_0 + 52X_0^2 = 270 + 52(X_0 - 37/104)^2 + 87/208 > 270.$$

This proves the curvature bound. The first-derivative comparison uses the polynomial increment

$$R(m, U) = \frac{\Pi_U(m, U, b_0) - \Pi_U(m, 0, b_0)}{U},$$

with its polynomial value at zero. Two fixed-chart quadratic lower models prove $R > 25000$; the exact intercept grouping gives $\Pi_U(m, 0, b_0) \geq -2985 + 503x + 12515x^2/2$. Substituting $b_0 = (1 - x^2)/8$ proves the stated restricted estimate. Grouping the U coefficients for $m \geq 1/3$ proves the bound 10700.

For clarity, the two physical-domain signs are not claimed on an unnecessary larger algebraic domain. Exact norm subtraction factors as

$$N_X - N_Y = \frac{8U(a - b)\rho^2}{E_a^2 E_b^2} \mathcal{H}(D, U).$$

The explicit power-grouping proof gives $\mathcal{H} \geq 1261/162$ on $0 \leq U \leq D^2/6$, $0 \leq D \leq 1$. Also the definition of Π has the weighted-conjugate identity

$$\Pi = \frac{(1 + D)^3 \mathcal{N}(D, e) + (1 - D)^3 \mathcal{N}(-D, e)}{4\rho^2}.$$

Each numerator is a constant minus two squares of affine functions of e , so $\Pi_{ee} < 0$ for the physical $0 < D < 1$. On $0 < U \leq r$, one has $K(m, U) < 1$: at $c = C_L(m)$, $K(m, 1 - c) - 1 = 4(c - 1)(c^3 + c^2 + c - 2) < 0$, and K increases in U .

All expansions, power groupings, model reconstruction identities and rational error bounds just specified are reproduced by the accompanying `verify_explicit_stage.py`. The complete identities and coefficient bounds, including the two increment models, are printed in its colocated mathematical supplement. This verifier uses no optimizer, no stored SOS witness, no subdivision search, and no floating-point sign test. These finite coefficient computations are part of the proof; they do not audit the preceding geometric forcing lemmas. $\square$

Lemma F.5 (Boundary estimates and coordinatewise monotonicity). *For $0 \leq m \leq 1/2$,*

$$\Pi(m, r, r) > \frac{1063}{1700}.$$

The affine-branch left endpoints obey

$$\Pi(m, 0, m) > 0 \quad (m \leq 1/3),$$

$$\Pi(m, U, 1/3) \geq 4096/27 \quad (m \geq 1/3, U \geq (m - 1/3)/3).$$

Moreover $\Pi_e \geq 1152$ on $\mathcal{R} \times [0, 1/3]$.

Proof. At $U = 0$, $x = 1 - 2m$, exact factorization gives

$$\Pi_e(m, 0, 1/3) = \frac{8}{3}(x^2 + 3)(21x^8 + 130x^6 + 289x^4 + 712x^2 + 144) \geq 1152,$$

and $\Pi_{ee}(m, 0, e) < 0$. The positive Π_{Ue} bound in lemma F.4 extends this to the whole rectangle. In particular Π is increasing in both U, e when $U \geq b_0$ and $b_0 \leq e \leq 1/3$.

Put $q(m) = -9m^2/10 + 98m/125 - 3711/100000$. The exact degree-eight comparison proves $b_0 \leq q \leq r$ for $1/5 \leq m \leq 1/2$. Thus

$$\Pi(m, r, r) \geq \begin{cases} \Pi(m, b_0, b_0) > 103, & m \leq 1/5, \\ \Pi(m, q, q) > 1063/1700, & m \geq 1/5. \end{cases}$$

The second polynomial has degree 27. With $x = m - 41/100$, its explicit Taylor coefficient bounds yield

$$\Pi(m, q, q) > 16/25 - 10|x| + 1700x^2 \geq 1063/1700.$$

The first interval and the comparison $q \leq r$ also use exact rational coefficient bounds, reproduced in the same standalone verifier and printed supplement. They are not consequences of numerical fitting. The parameter q is only an auxiliary boundary comparison; it does not replace r , e_0 , or the actual disk.

For $m \leq 1/3$, the factor $(2m-1)^2$ can be removed from $\Pi(m, 0, m)$, and grouping higher negative powers under lower positive powers on $0 \leq 1-3m \leq 1$ proves positivity. For $m \geq 1/3$, put $x = 1-2m \in [0, 1/3]$. Direct expansion gives $\Pi(m, 0, 1/3) \geq -96 + 6688x^2/3$. Integrating $\Pi_U(m, U, 1/3) > 10700$ to $U \geq (1-3x)/18$ gives a decreasing quadratic whose minimum is $4096/27$. $\square$

Theorem F.6 (Explicit disk-tangent certificate). *On the strict F domain, if $e_0 \leq 1/3$, then*

$$Q_X(e_0) > 0, \quad Q_Y(e_0) > 0.$$

The assertion uses the fixed-start comparison points, not the old junction-start Newton points.

Proof. Concavity of $\mathcal{B}$ and its endpoint bounds give $\mathcal{B}(m, U, e) > 4$ on $b_0 \leq e \leq m$, hence at $e = e_0$. The elementary identity

$$D(1+3D^2) - D^2(3+D^2) = D(1-D)^3 \geq 0$$

therefore gives

$$\mathcal{A} + D\mathcal{B} \geq \frac{\Pi}{1+3K}.$$

It suffices to prove $\Pi(m, U, e_0) > 0$.

For $U \geq r$, $e_0 = r$; the restricted derivative estimate and $\Pi_{Ue} > 0$ give $\Pi_U(m, U, r) > 0$. The boundary estimate at $U = r$ settles this range. For $U \leq r$, put $f(U) = \Pi(m, U, m - \kappa U)$ and $S_k(e) = \Pi_{UU} - 2k\Pi_{Ue} + k^2\Pi_{ee}$. For $0 \leq e \leq 1/3$ and the actual $\kappa \geq 1$,

$$\partial_e S_\kappa = (1-3e)H_2 + 3eH_3 + 2(\kappa-1)H_1 > 0.$$

On the integration path, $\Pi_{ee} < 0$ and $\Pi_{Ue} > 0$, so $\partial_k S_k < 0$ for $k \geq 0$. Consequently

$$f''(U) = S_\kappa(e) \leq S_\kappa(1/3) \leq S_{\ell(m)}(1/3) < -21500.$$

The right endpoint is $U = r$. The left endpoint is zero if $m \leq 1/3$; otherwise it is $U_3 = (m - 1/3)/\kappa \geq (m - 1/3)/3$. All endpoint values are positive by lemma F.5. Concavity proves positivity throughout the interval.

Thus $Q_X(e_0) > 0$. Exact subtraction gives

$$Q_Y(e) - Q_X(e) = (1-e)(1-3e)(N_X - N_Y) \geq 0 \quad (0 \leq e \leq 1/3),$$

using the established norm ordering. This proves both signs. $\square$